SAP Analytics Cloud - Story Builder

This case study explains the process of creating a sales dashboard in SAP Analytics Cloud (SAC). The individual work steps explain the structure of a story in the Story Builder.

Product

SAP Analytics Cloud
Global Bike

Level

Bachelor's Degree
Master
Beginner

Focus

Business Intelligence
Data Visualization

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Version

3.0

Last Change

March 2025

MOTIVATION

An example sales dashboard was demonstrated in the exercises for SAP Analytics Cloud.

In this case study, you work out a sales dashboard yourself in the SAC Story Builder. The required activities include the creation of a story as well as further settings options for the SAC-specific elements.

PREREQUISITES

Before you edit the case study, you should familiarize yourself with the navigation in SAP Analytics Cloud.

REMARK

This case study uses a model with historical sales data from the fictitious company Global Bike.

THANK YOU

This case study was developed with the help of students and employees of the Pforzheim University of Applied Sciences as well as scientific and student assistants of the SAP UCC Magdeburg.

SAP University Alliances

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Process Overview

Creating a Sales Dashboard as a Story in SAP Analytics Cloud (SAC) **learning objective**.

Time 220 min.

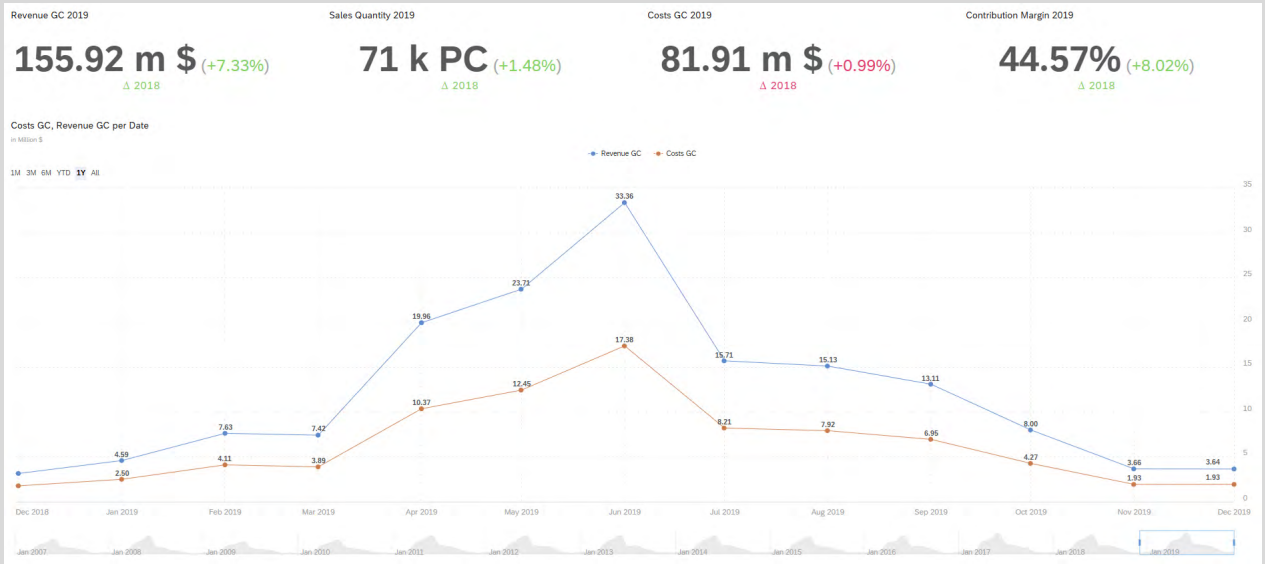
Scenario In her weekly video conference, Alex Kuhn (vice president of marketing and sales) informs his colleagues Andrea Marino (sales manager USA) and Hanna Ortmann (head of sales Germany) that the group has now also licensed the *SAP Analytics Cloud (SAC)* cloud solution in addition to SAP *Data Warehouse Cloud*. He orders the two to deal with SAC and set up a sales dashboard for Global Bike. It gives them the specifications mentioned below.

Specifications Alex Kuhn: "I've heard that in an SAC story, you can structure the charts into different pages or tabs. I suggest the following pages:

- Overview
- Customer View (3 Years)
- Product View
- Most Popular Bikes 2019
- Dynamic Analysis

"On the Overview page, I want to get a quick overview. I would like to show the most important key figures Revenue GC, Sales Quantity, Costs GC, and Contribution Margin for 2019 with comparative figures for the previous year.."

"It should also be possible to evaluate the key figures Revenue GC and Costs GC over the entire time period of the data."



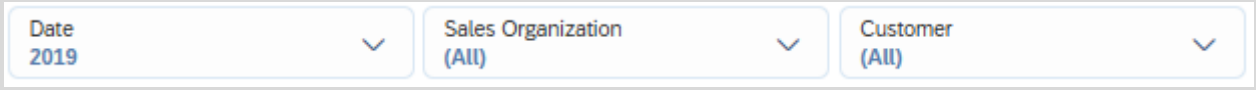
For the *customer view* of the last three years, I first imagine a table:

Customer/ Product View

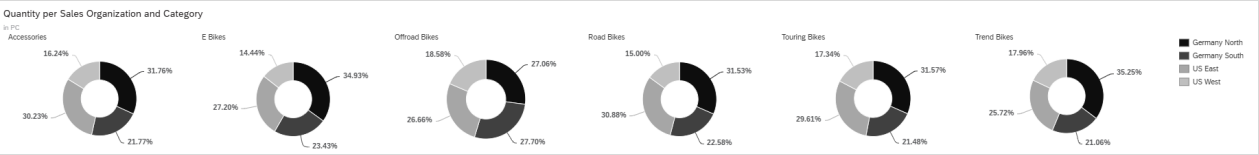
Date	Customer	Product	Measures	Revenue GC	Sales Quantity	Costs GC	% Rev Accessories
2019	(all)	(all)		155,915,470.89 $	71,343 PC	81,910,953.42 $	0.76%
	DE	(all)		90,121,269.91 $	38,460 PC	44,533,991.03 $	0.75%
	Germany North	(all)		50,782,009.38 $	21,977 PC	25,152,484.65 $	0.80%
	Germany South	(all)		39,339,260.53 $	16,483 PC	19,381,506.38 $	0.69%
	US	(all)		65,794,200.98 $	32,883 PC	37,376,962.39 $	0.77%
2018	(all)	(all)		145,269,843.27 $	70,302 PC	81,108,230.65 $	0.76%
	DE	(all)		83,695,373.06 $	38,927 PC	45,179,783.40 $	0.75%
	Germany North	(all)		46,901,756.88 $	22,016 PC	25,343,521.71 $	0.76%
	Germany South	(all)		36,793,616.18 $	16,911 PC	19,836,261.69 $	0.73%
	US	(all)		61,574,470.21 $	31,375 PC	35,928,447.25 $	0.78%
2017	(all)	(all)		152,345,747.48 $	70,599 PC	81,825,384.92 $	0.73%
	DE	(all)		87,333,547.08 $	37,588 PC	43,610,003.43 $	0.73%
	Germany North	(all)		46,604,430.38 $	20,316 PC	23,331,033.25 $	0.75%
	Germany South	(all)		40,729,116.70 $	17,272 PC	20,278,970.18 $	0.70%
	US	(all)		65,012,200.40 $	33,011 PC	38,215,381.49 $	0.73%

“You may already be able to display additional overviews of sales organizations there.“

“On the *Product View* page, I’d like an analysis with the *sales quantity* and *contribution margin*. It should be possible to filter the entire page by *Date*, *Sales Organization*, *and Customer*.“



“The percentage distribution of the *sales quantity* by *sales organization* per *product category* should be easy to evaluate.“



“Mrs Marino, Mrs Ortmann, you know that our contribution margin reaches a critical point at 46%. If it is possible to set up a threshold value for this value and evaluate it graphically, for example:

Contribution Margin GC	Contribution Margin %
68,826,302 $	44.48%
68,826,302 $	44.48%
68,826,302 $	44.48%
3,475,673 $	39.27%
18,276,925 $	47.09%
19,885,494 $	45.53%
26,843,992 $	42.92%
344,217 $	40.77%

"In addition, the monthly quantities of the *Bicycles division* are to be displayed in a *heat map*."

Quantity for Bicycles

in PC

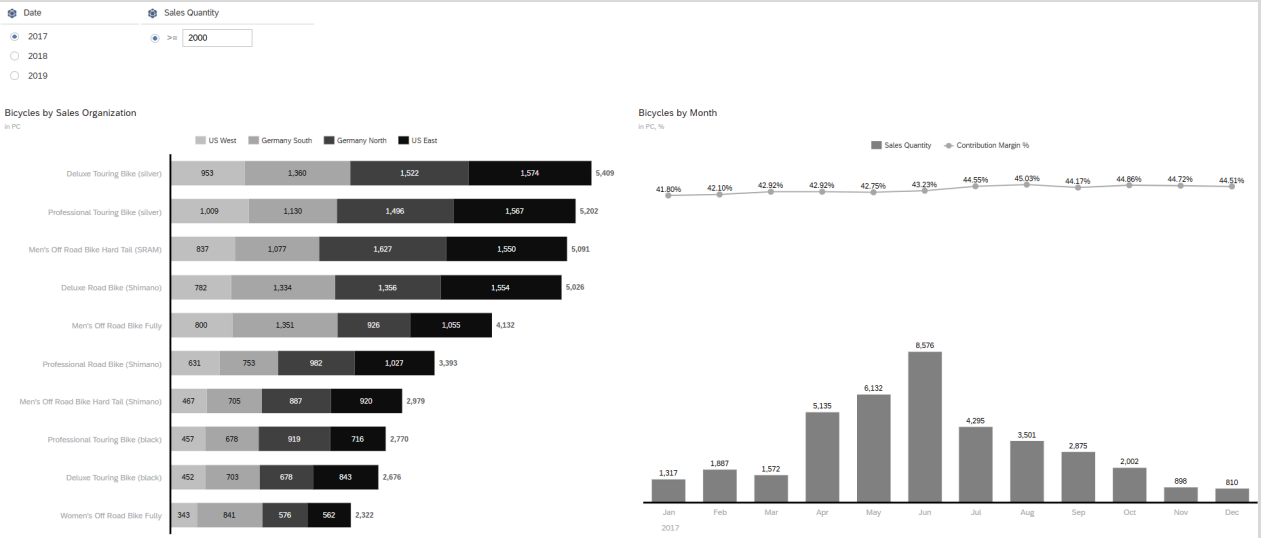
	E	Offroad	Road	Touring	Trend
Jan	39	359	340	584	5
Feb	67	837	603	761	20
Mar	69	696	581	750	10
Apr	178	1,874	1,715	1,831	67
May	330	2,213	1,874	2,183	73
Jun	368	3,170	2,485	3,364	85
Jul	164	1,457	1,193	1,656	5
Aug	127	1,469	1,212	1,401	61
Sep	149	1,097	937	1,467	44
Oct	83	737	685	785	20
Nov	30	409	310	310	17
Dec	65	410	295	273	44

"I also want a presentation of the five most frequently sold products and the five most rarely sold products with previous year's comparisons."

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"On the *Most Popular Bikes* page, I want to see all the bikes of which we sold more than 2000 pieces."



"It would be nice if I can select individual elements in one of the two graphics and then filter on this selection in the second graphic."

"Finally, I want a *dynamic analysis* of various measures and dimensions of the last five years."

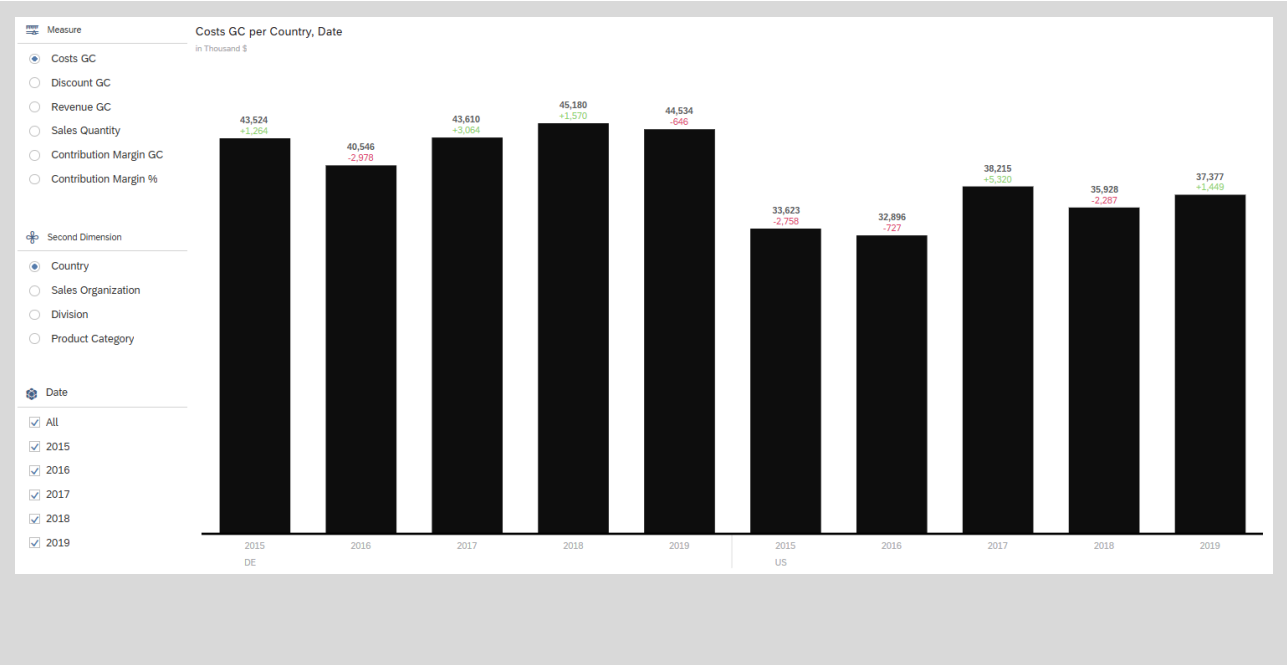

Measure
Costs GC
Discount GC
Revenue GC
Sales Quantity
Contribution Margin GC
Contribution Margin %
Second Dimension
Country
Sales Organization
Division
Product Category
Date
All
2015
2016
2017
2018
2019
Costs GC per Country, Date
in Thousand $
43,524
+1,264
40,546
-2,978
43,610
+3,064
45,180
+1,570
44,534
-646
33,623
-2,758
32,896
-727
38,215
+5,320
35,928
-2,287
37,377
+1,449
2015
2016
2017
2018
2019
DE
2015
2016
2017
2018
2019
US

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Step 1: Log on to SAC

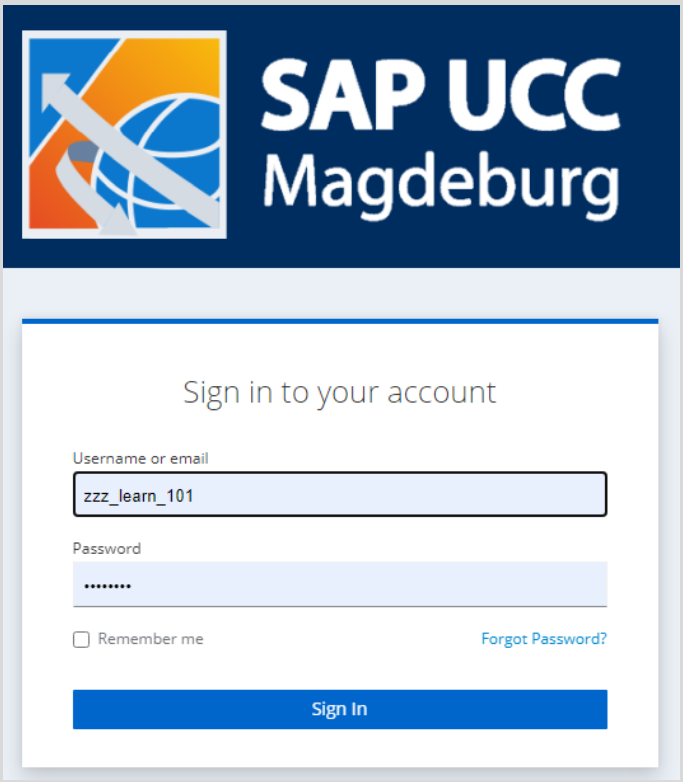
Task Log on to SAP Analytics Cloud. **Time** 10 min.

Description Log on to the SAC tenant in a web browser.

Initial Screen

Clicking on the following link takes you to the Identity Management System of SAP UCC Magdeburg. This provides anonymous identities for logging on to SAP cloud solutions according to valid data protection rights.

https://ucc-pilot-sac.eu10.hcs.cloud.sap/



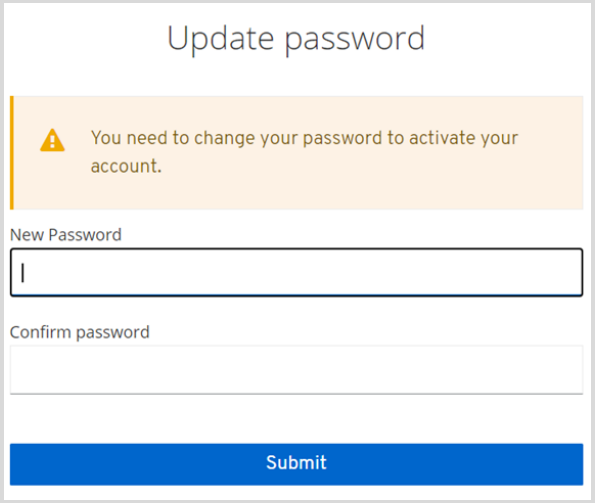
User
Password
Sign In

On the login screen, enter the **user** and **password** provided by your instructor. Then click **Sign In**.

Note If you log on to SAC with this user for the first time and with an initial password, you are prompted to enter a personal password for this user.

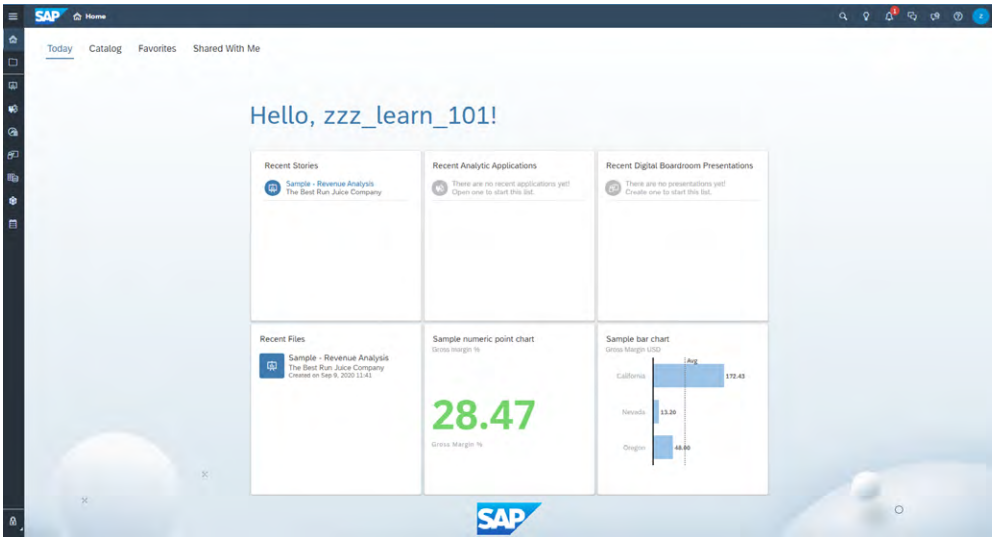
Password of your choice

After you have entered your new password twice in the *Update password* window, confirm with **Submit**.



After logging on, you will be taken to the home *screen*.

Home Screen



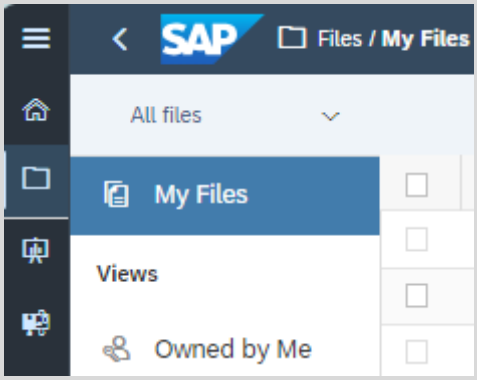
Step 2: View the Analytic Model

Task Familiarize yourself with the underlying model. **Time** 15 min.

Description Before you start a story in SAC to visualize the new sales dashboard, start by looking at the analytic model that IT colleagues have created in advance.

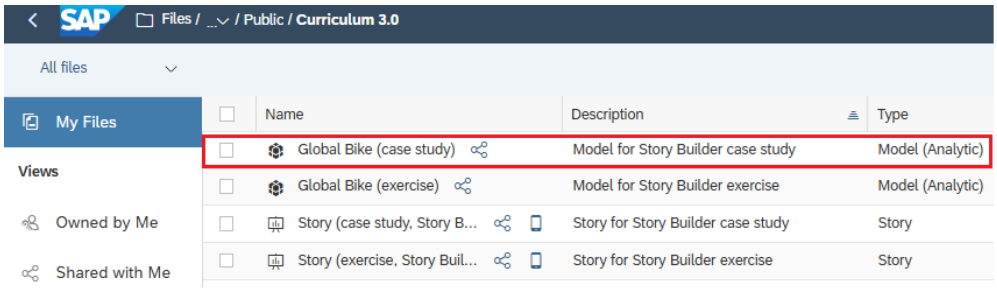
Initial Screen

On the home screen, click (*Files*) in the upper left corner.



Public ->Curriculum 3.0
Global Bike (case study)

Navigate to the folder **Public ►Curriculum 3.0**. Here you select the model **Global Bike (case study)**.



Model

A *model* represents the business data of an organization or business segment. You can use a model as a basis for your story.

Model Types

There are two *types of models* available in SAP Analytics Cloud:

- *Imported*: Data is imported (copied) and stored in SAP Analytics Cloud. Changes in the source system do not affect the imported data.
- *Live*: data is stored in the source system. They are not copied to SAP Analytics Cloud, meaning that any changes to the source data are immediately available unless structural changes are made to the table or SQL views.

Model Types

In addition, there are two types of models:

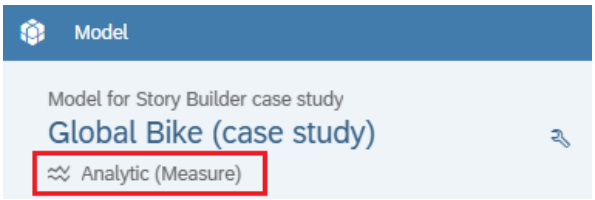
- *Analytic Model*: These are fully functional models that contain business data. It is the more flexible model type. Typically, analytic

models are used to analyze data and uncover trends and anomalies. Analytic models can be consumed with live data (see above).

- *Planning model*: Planning models are used to support business planning tasks, e.g. B. Forecasts,, and used in the optimization of the planning process. Planning models cannot be created from a live data source.

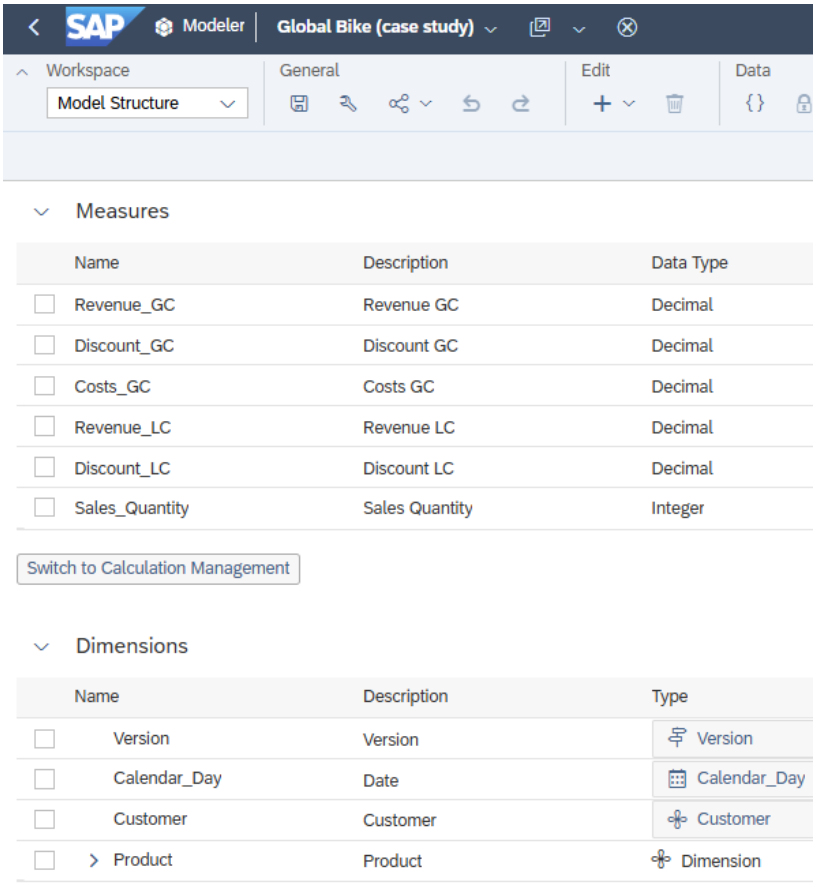
As the name of the *Global Bike (case study)* model suggests, the model used in this case study is a new analytic model and focuses on the evaluation of defined key figures.

On the right-hand side of the screen in the *Model Details*, you can see that it is a measure-based analytic model.



Note If you do not see the model details, click Details in the upper right corner.

You see a list of six measures and four dimensions of this model.



Note Models in financial and business reporting can have different measures that are reported in both global currency (GC) and local currency (LC). Revenue GC (Global Currency) and Revenue LC (Local Currency) are terms used in financial reporting and in the international business environment. They refer to the display of sales in different currencies.

Revenue GC (Global Currency): Refers to the recording and reporting of sales in a single, global currency, often referred to as consolidation currency. This currency is used to standardize the financial performance of a multinational company so that financial information is comparable across different countries and regions. A strong and stable currency such as the US dollar or the euro is often used for this purpose. The conversion is done using exchange rates to transfer local sales to this global currency.

Revenue LC (Local Currency): This key figure shows the revenue as recorded in the respective national currency of the location where the company operates. Sales figures in local currency provide an accurate view of financial performance in a specific market or country without the influence of exchange rate fluctuations.

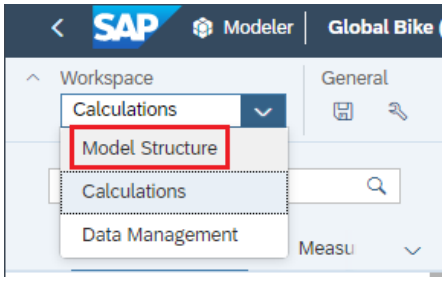
By using Revenue GC and Revenue LC, companies can both analyze local market performance and get a consolidated view of the entire organization, which is essential for strategic decisions and reporting.

You can also choose Switch to Calculation Management to display the calculated key figure *ContributionMargin USD*. As you can see in the formula above, it is calculated as follows:



To return to the graphical model structure, choose Model Structure from the dropdown menu at the top left.

Model Structure



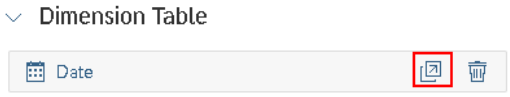
To get started, click the **Calendar_Day** time dimension to view the *Settings dimension* on the right.

Calendar_Day

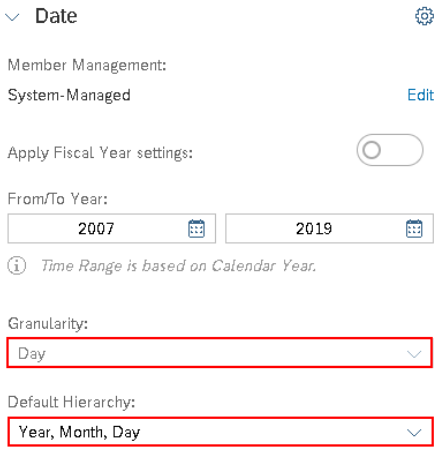
Note If you do not see the settings, click Details in the upper right corner.

Edit Dimension Table

Under *Dimension Table*, click **Edit Dimension Table** ( ) to see the settings in detail.

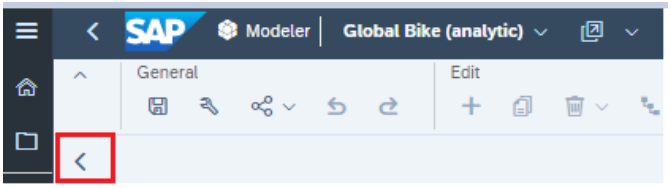


In the *Settings dimension*, you can see that *day* (day level) has been selected as the granularity for this time dimension. *Year*, *Month*, and *Day* are also defined as hierarchy levels.



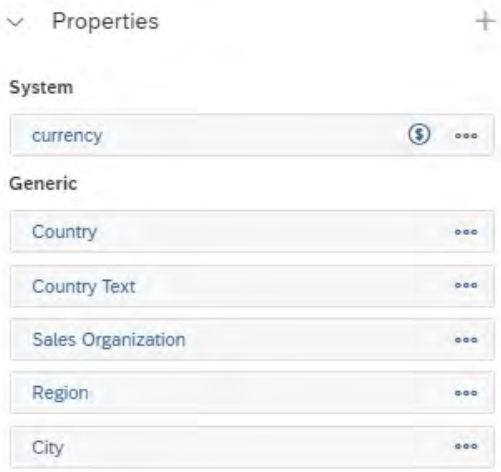
Back

To return to the dimension view, choose **Back** in the header area.



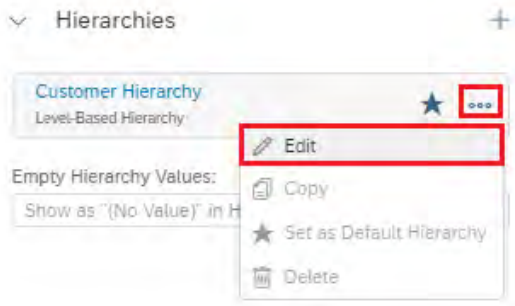
Customer
Edit Dimension Table

Next, click the **Customer** dimension in the model and choose **Edit Dimension Table** under *Dimension*. If you scroll to the bottom right of the Settings dimension, you can see the six available attributes of this dimension in the *Properties* area.



Below this, you find the *Hierarchies* section. From there, click **More** and then **Edit** next to *Customer Hierarchy*.

More
Edit

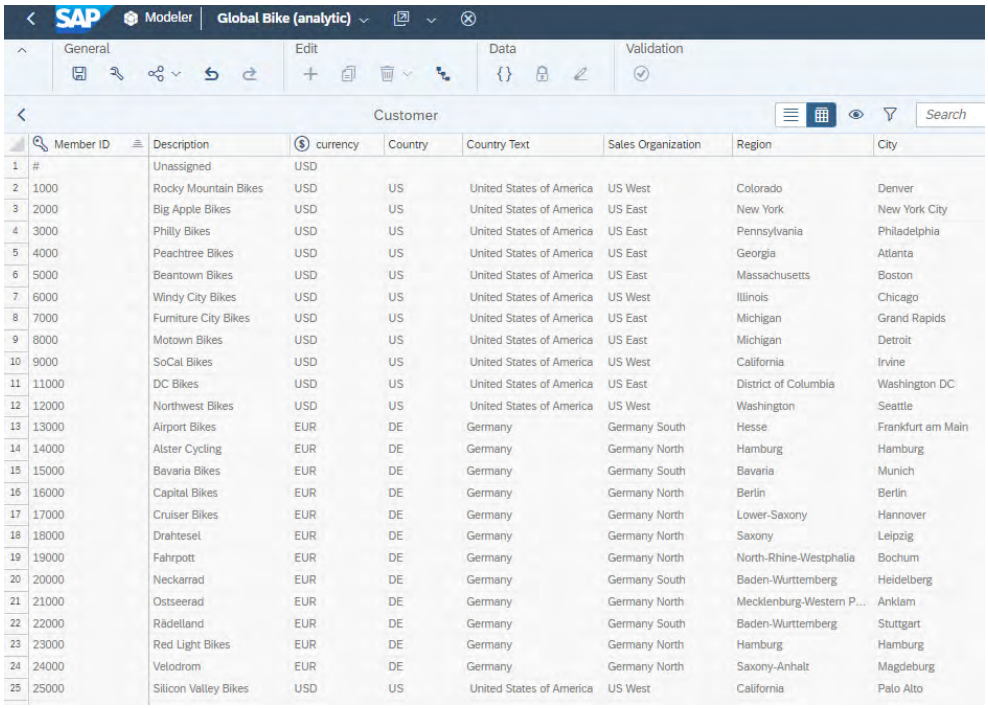


The system displays the defined customer hierarchy with the levels *Country*, *Sales Organization*, *Region*, and *Member ID*.



Click **OK** to close the *Hierarchy Builder*. You now see all 24 customers of Global Bike in a list with attributes.

Ok then



	Member ID	Description	currency	Country	Country Text	Sales Organization	Region	City
1	#	Unassigned	USD					
2	1000	Rocky Mountain Bikes	USD	US	United States of America	US West	Colorado	Denver
3	2000	Big Apple Bikes	USD	US	United States of America	US East	New York	New York City
4	3000	Philly Bikes	USD	US	United States of America	US East	Pennsylvania	Philadelphia
5	4000	Peachtree Bikes	USD	US	United States of America	US East	Georgia	Atlanta
6	5000	Beantown Bikes	USD	US	United States of America	US East	Massachusetts	Boston
7	6000	Windy City Bikes	USD	US	United States of America	US West	Illinois	Chicago
8	7000	Furniture City Bikes	USD	US	United States of America	US East	Michigan	Grand Rapids
9	8000	Motown Bikes	USD	US	United States of America	US East	Michigan	Detroit
10	9000	SoCal Bikes	USD	US	United States of America	US West	California	Irvine
11	11000	DC Bikes	USD	US	United States of America	US East	District of Columbia	Washington DC
12	12000	Northwest Bikes	USD	US	United States of America	US West	Washington	Seattle
13	13000	Airport Bikes	EUR	DE	Germany	Germany South	Hesse	Frankfurt am Main
14	14000	Alster Cycling	EUR	DE	Germany	Germany North	Hamburg	Hamburg
15	15000	Bavaria Bikes	EUR	DE	Germany	Germany South	Bavaria	Munich
16	16000	Capital Bikes	EUR	DE	Germany	Germany North	Berlin	Berlin
17	17000	Cruiser Bikes	EUR	DE	Germany	Germany North	Lower-Saxony	Hannover
18	18000	Drahtesel	EUR	DE	Germany	Germany North	Saxony	Leipzig
19	19000	Fahrpott	EUR	DE	Germany	Germany North	North-Rhine-Westphalia	Bochum
20	20000	Neckarrad	EUR	DE	Germany	Germany South	Baden-Wurttemberg	Heidelberg
21	21000	Ostseerad	EUR	DE	Germany	Germany North	Mecklenburg-Western P...	Anklam
22	22000	Rädelland	EUR	DE	Germany	Germany South	Baden-Wurttemberg	Stuttgart
23	23000	Red Light Bikes	EUR	DE	Germany	Germany North	Hamburg	Hamburg
24	24000	Velodrom	EUR	DE	Germany	Germany North	Saxony-Anhalt	Magdeburg
25	25000	Silicon Valley Bikes	USD	US	United States of America	US West	California	Palo Alto

Click **Back**  to return to the *Model Structure*.

Back

Finally, click the **Product** dimension and look at the available attributes, the hierarchy, and other details.

Product

Which hierarchy levels are defined for this dimension?

Level 1: _____

Level 2: _____

Level 3: _____

How many products does the table contain? _____

You have now understood the structure of the model used in this case study and can start creating your story.

Step 3: Create a new story

Task Open SAP Analytics Cloud and create a new story. **Time** 10 min.

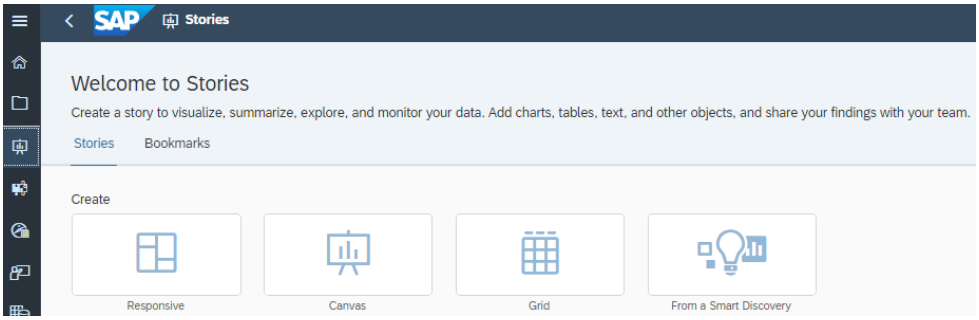
Description Open your SAC tenant and log on. You then create a new story based on an existing model.

Initial Screen

To go to the home screen, click **Home**  .

Story

To create a new story, choose **Stories**  .



Layouts

You can choose from four different layouts for your new story:

- The *responsive* page allows you to create pages that can be accessed on all mobile devices due to their dynamic adjustment. However, as a result, it is not possible to create charts on top of each other, for example.
- The *canvas* page allows elements (charts and tables) to be freely arranged on pages. In this form of pages, you can, for example, place elements on top of each other or mix them together. In contrast to a *responsive* page, the elements on different end devices do not adapt automatically.
- The *grid* page only supports the display of tables. The page form enables, for example, additional calculations for cells, similar to the spreadsheet MS Excel.
- *With From a Smart Discovery*, a data model can be examined for correlations automatically and using predefined automatisms. In this way, corresponding charts and tables can be generated, which explain and display the main influencing factors and data connections, supplemented by corresponding texts for explanation.

The *Grid* Page and *From a Smart Discovery* are no longer supported in the *Optimized Design Experience* and are therefore not part of this exercise.

As part of the creation of your story, you will learn about *responsive* and *canvas* layouts.

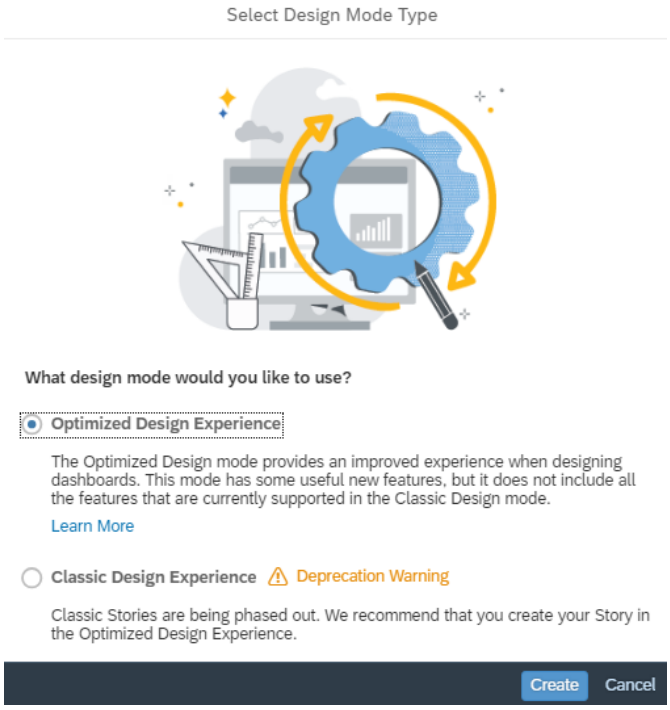
Responsive

First, start with a responsive page. To do this, click **Responsive**.



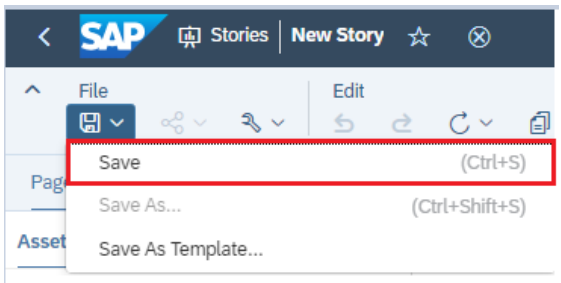
In the window that opens, select the *Optimized Design Experience* option and click **Create**.

Create



Before proceeding, save the story to your folder. To do this, click the *Save* icon in the *File* area of the ribbon and choose **Save**.

Save

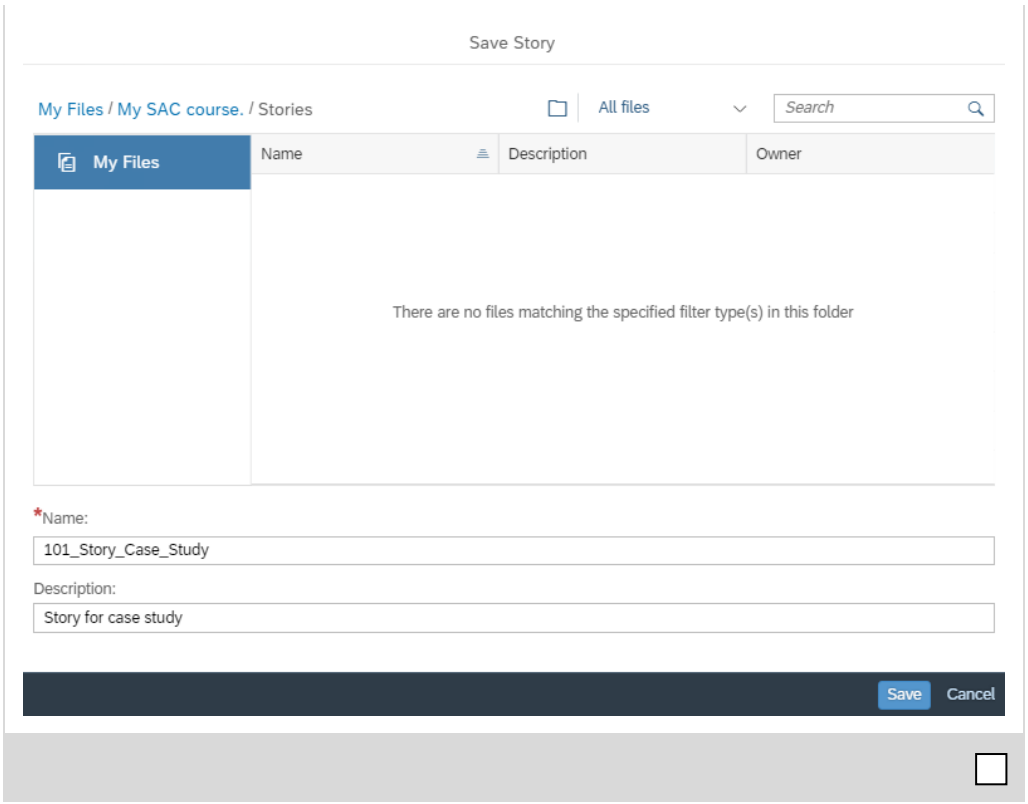


First, select the *Stories* subfolder as the location in your personal folder. You can access this via *My Files* ► *My SAC course* ► *Stories.*

Then enter **###_Story_Case_Study** in the *Name* field and **Story for case study** in the *Description* field. Note that you replace ### with your three-digit number provided by your instructor. Compare your entries with the following screenshot and confirm the window with **OK**.

###_Story_Case_Study
Story for Case Study

Ok then


Save Story
My Files / My SAC course. / Stories
All files
Search
My Files
Name
Description
Owner
There are no files matching the specified filter type(s) in this folder
*Name:
101_Story_Case_Study
Description:
Story for case study
Save
Cancel

Step 4: Create a graphic (time series)

Task Create a graphic of type Time Series based on the *Global Bike (analytic)* model.

Time 15 min.

Description Define the graphic of the time series type and adjust the display.

Initial Screen

You have still opened the *Story_Case_Study* story from the previous step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Graphic

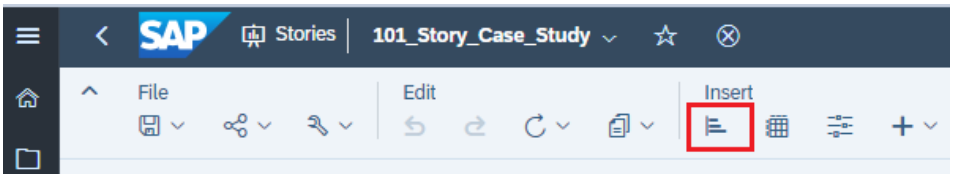
Graphics provide great ways to better visualize your data and present the corresponding key statements in a more conclusive way.

The SAC story offers you a variety of different graphic types. Some of these graphic types will be presented in detail during the course of this case study.

Start with the *Time Series* graphic type to visualize sales and costs across the entire dataset.

Chart

To insert a graphic into your story, choose the **Chart** icon in the *Insert* area of the ribbon.



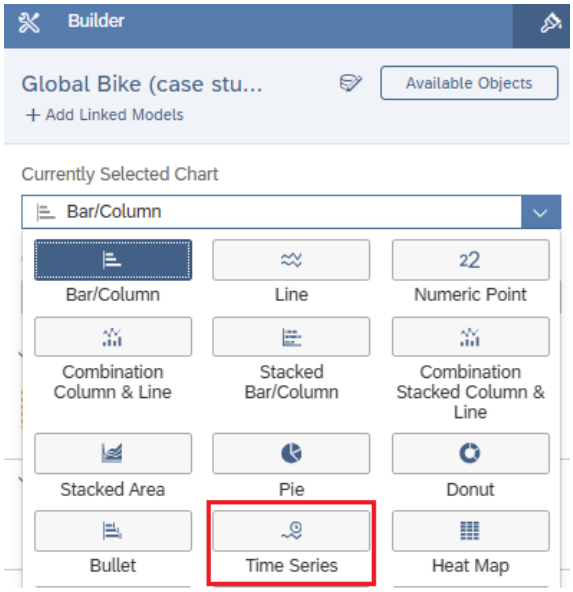
Public -> Curricula 3.0 Global Bike (case study)

First, you must define which data source is to be used. Open the folder **Public ►Curricula 3.0** and select the **Model Global Bike (case study)**.

After you assign the model, the *right-hand side panel* opens in the *Builder* panel.

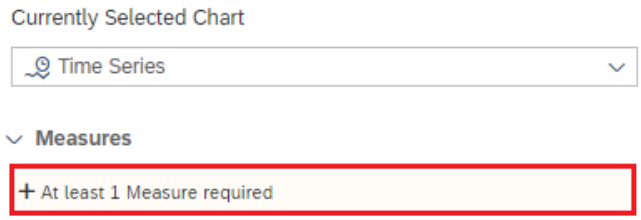
Current Selected Chart Time Series

Change the graphic type to *Time Series*. To do so, open the dropdown under **Current Selected Chart** and select the **Time Series** entry.



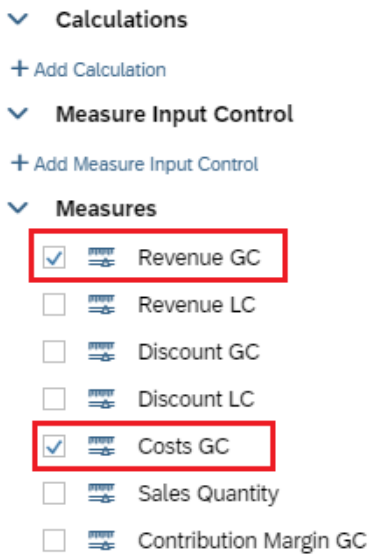
In the *Measures* section, choose **At least 1 Measure required**.

At least 1 Measure required



Now select the key figures **Revenue GC** and **Costs GC** one after the other. Please compare with the screenshot before proceeding.

Revenue GC
Costs GC



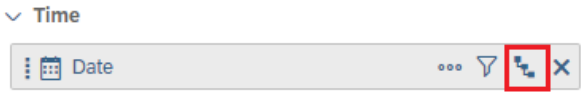
Click in the blue line of the *right-hand side panel* to return to the definition of the graphic. Then define the time dimension of your graphic.

In the *Time* section, click the **At least 1 Dimension required** entry and select **Date**.

At least 1 dimension required
Date

To change the initial display from day to month, choose *Date* under *Time* and choose **Hierarchy** .

Hierarchy



Choose the **Month** hierarchy level.

Month

In the *right-hand side panel*, switch from the *Builder* panel to the *Styling* panel. To do so, click **Styling** .

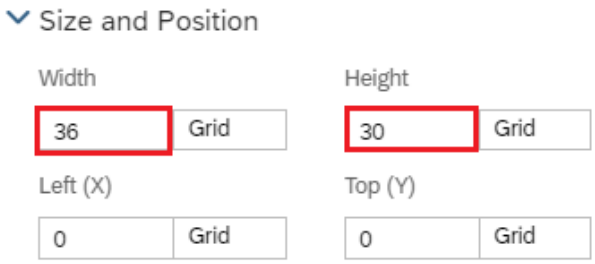
Styling



Under *Size and Position*, change *Widht* to **36 Grid** and *Height* to **30 Grid**.

36 Grid
30 Grid

Time -> Show Totals



Now scroll down to the *Number Format* area. Under *Scale*, select The value **Million** and the data in millions.

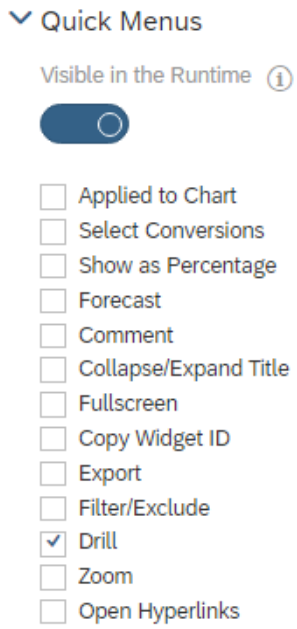
Million



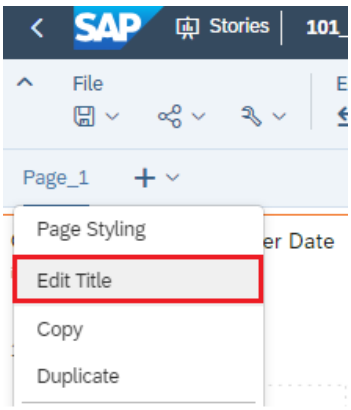
In view mode, it should only be possible to change the drill of time.

For this setting, scroll up again to the *Quick Menus* area in the *right-hand side* panel. Select all options here except **Drill**.

Drill



Rename the title of the first page to *Overview*. To do so, click the arrow next to Page 1 and select **Edit Title**.



Finally, in the menu bar in the *File* area, choose Save icon again and then **Save** to save your changed story.

Save

Step 5: Structuring the Story

Task Structure your story through a second tab. **Time** 10 min.

Description Create a second tab in your story and give it a meaningful name.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Pages

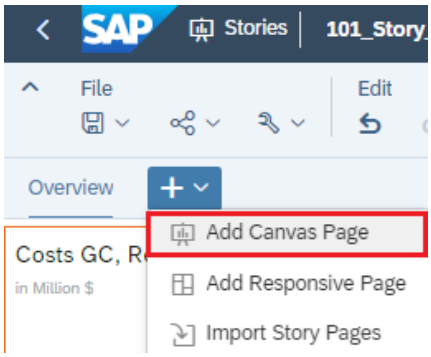
To better structure your SAC story, you have the option to create multiple *pages* in your story. This procedure also makes it easier for the end user to use the story.

Within a story, you can specify for each tab whether it is a *canvas page* or a *responsive page*. For better usability and consistency, you should generally ensure that you use only one page type within your story, if possible.

This case study intentionally deviates from this procedure so that you can get to know the two page types in practice.

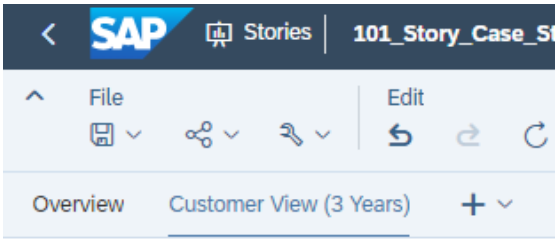
Canvas

Currently, your story includes a page called *Overview*. To add another page, go to the area next to the *Overview* page, choose *Add New Page* and select the **Add Canvas Page** entry.



Customer View (3 Years)

Now you get another page, which you rename to **Customer View (3 Years)**. The navigation bar of your story should now look like this.

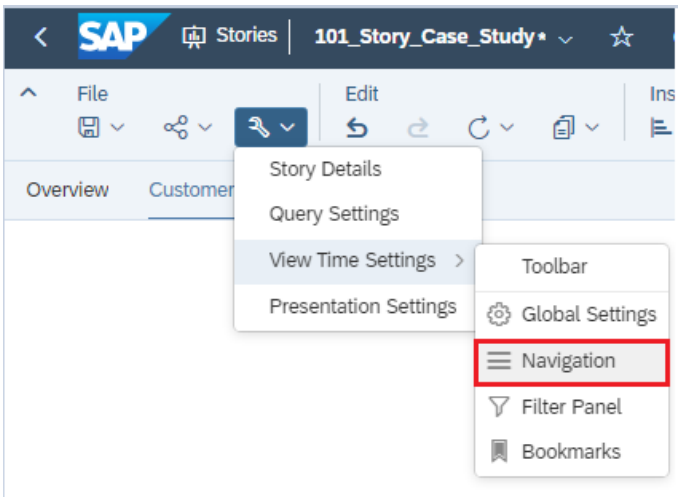


For a story with multiple pages, it should be possible to switch pages using tabs for easy navigation.

Check these settings for easy navigation for your story and make adjustments if necessary.

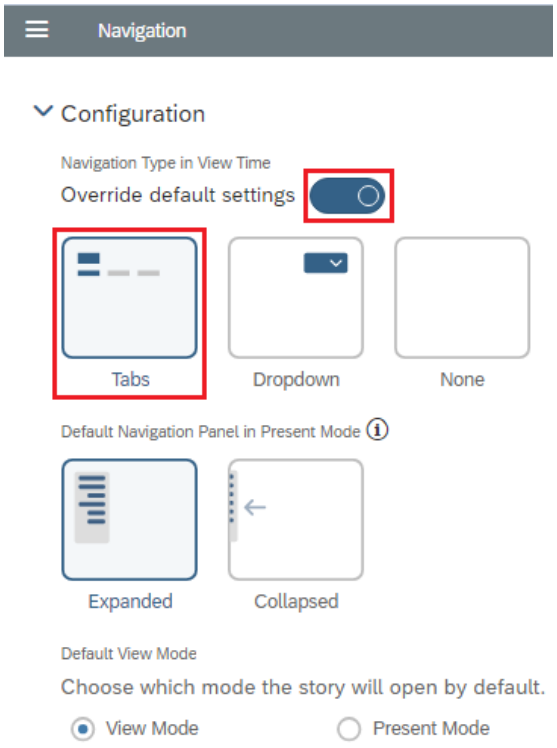
In the *File* area of the ribbon, choose The Icon **Story Settings** and choose **View Time Settings** ► **Navigation**.

Story Settings
View Time Settings →
Navigation



If *Tabs* is not selected under *Configuration* in the right *navigation pane*, use **Override default settings** to change the setting and select the **Tabs** option. Confirm your customizations with **Done**.

Override Default Settings
Tabs
Done



Finally, in the menu bar in the *File* area, choose *Save* icon again and then **Save** to save your changed story.

Save

Step 6: Create a table

Task Create a table with an individual analysis view on the existing data basis. **Time** 15 min.

Description On the *Customer View (3 Years)* page, define the rows and columns of the table and adjust their appearance.

Initial Screen

You have still opened the story *Story Case Study* from the previous step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

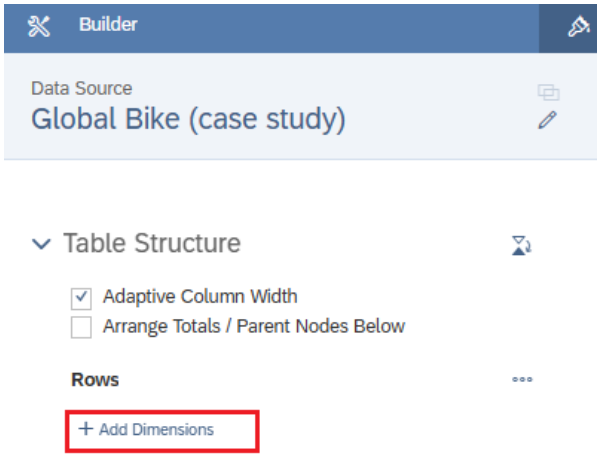
Table

In the *Insert* area of the ribbon, choose the Icon **Table** .



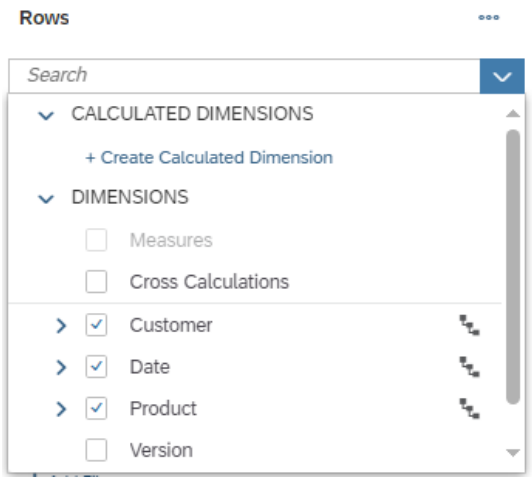
Add Dimensions

In the right-hand side panel that opens, first make sure that the *Builder* settings are displayed. Then, in the *Rows* section, choose **Add Dimensions**.



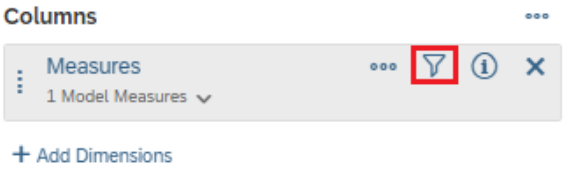
Date
Customer
Product

Now select the dimensions **Date**, **Customer, and Product** one after the other. Please compare with the screenshot before proceeding.



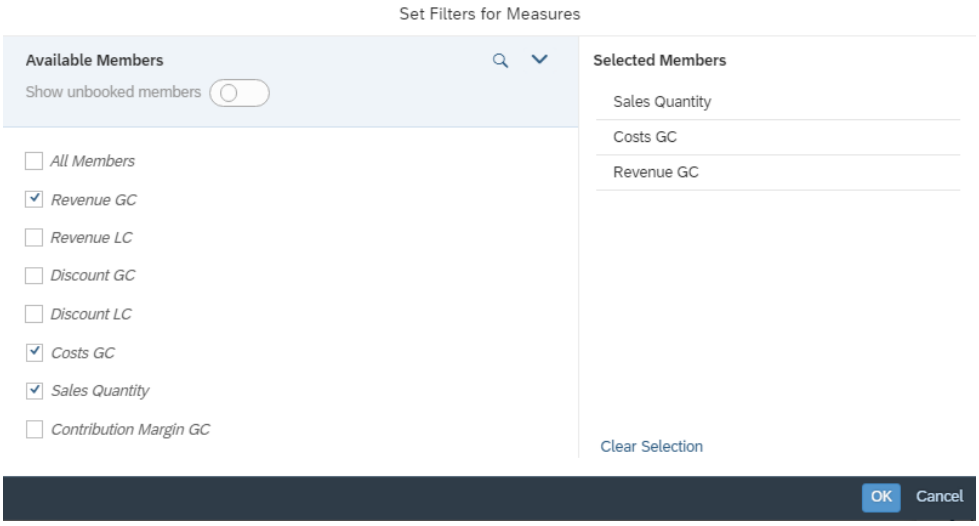
You now define the key figures to be displayed in your table. To do so, move your mouse to the *Columns* section and click **Manage Filters** next to *Measures*.

Manage Filters



In the window that opens, select the key figures **Sales Quantity**, **Cost GC**, and **Revenue GC** and confirm the window with **OK**.

Sales Quantity
Cost GC
Revenue GC
Ok then



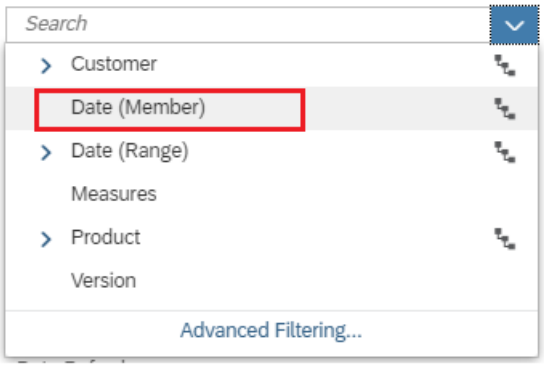
Since you only want to view the data from the last three years in the tabular evaluation, choose **Add Filters** in the *Filter* area.

Add Filters



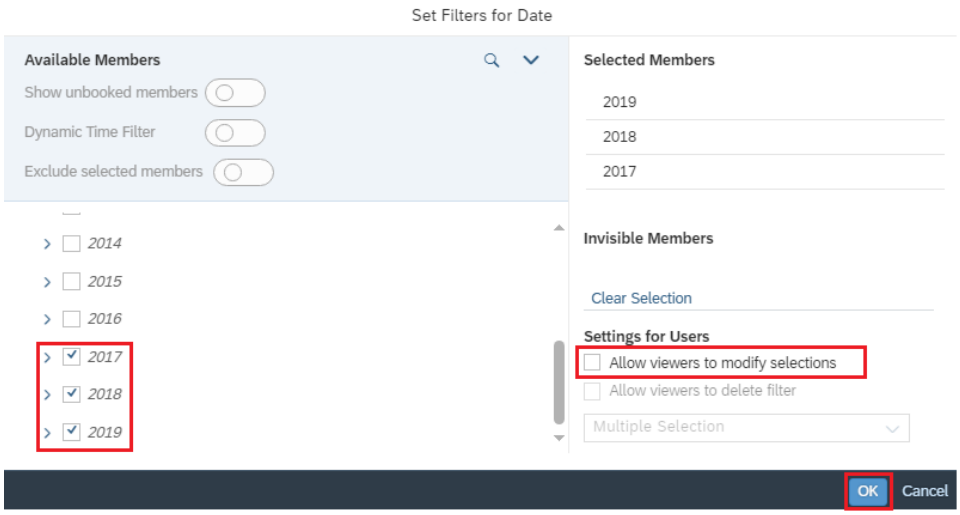
Select **Date (Member)** and select the last three years.

Date (Member)



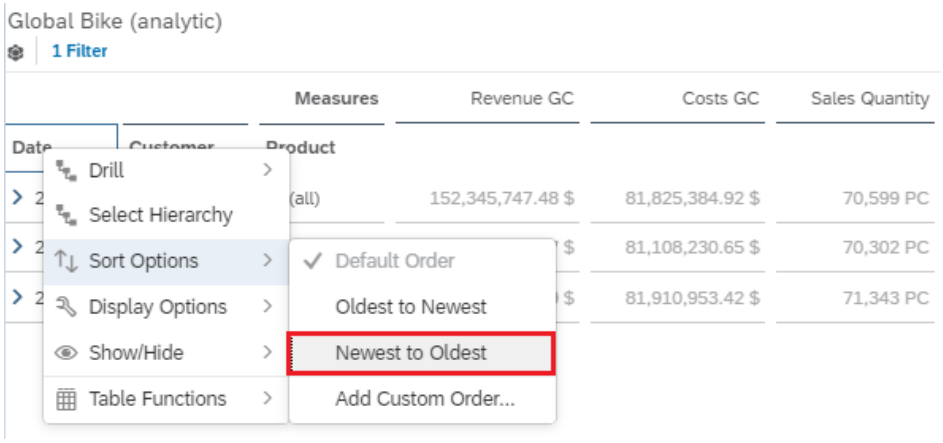
Since this should be a rigid filter, that is, a fixed filter that cannot be changed by the end users, deselect **Allow viewers to modify selections** and confirm the window with **OK**.

~~Allow viewers to modify selections~~
Ok then



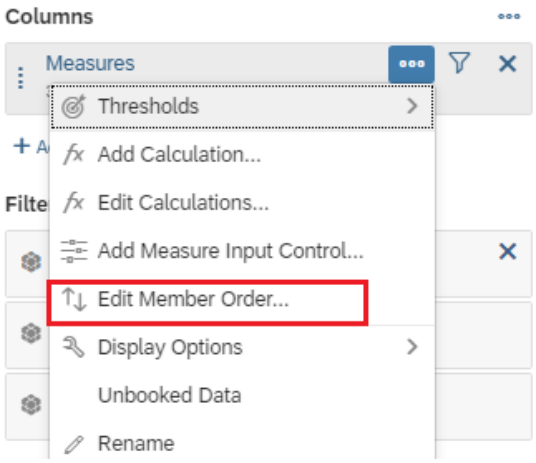
Since you want to see the most recent data first, the *Date* dimension must be sorted in descending order. To do so, right-click **Time** ► **Sort Options** ► **Newest to Oldest** in the table.

Date ->
Sort Options ->
Newest to Oldest



More
Edit Member Order

For better readability, the key figures should be displayed in the sequence *Revenue GC*, *Sales Quantity*, and *Cost GC*. To do this, click on the **More** icon on the right side of the *Builder* under *Measures* and choose **Edit Member Order**.



Revenue GC
Sales Quantity
Cost GC
Done

Now use drag and drop to change the sequence to the above and confirm the view at the bottom right with **Done**.



Save

In the menu bar in the *File* area, choose **Save** and then **Save** to save your changed story.

Customer/Product View

Adjust the header of the table. To do this, double-click in the text and enter customer/**product view** instead of the model name.

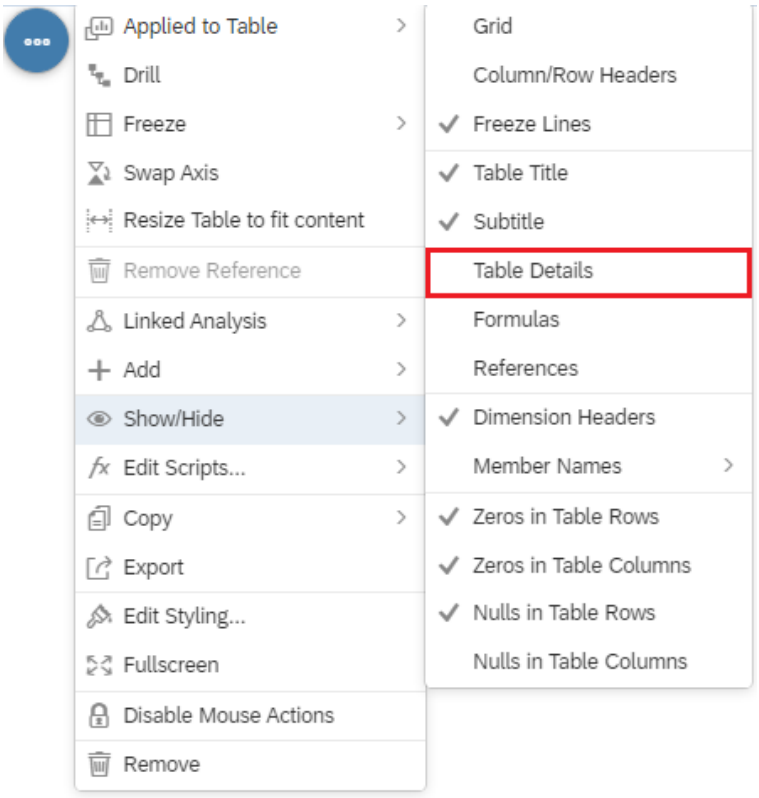
Customer/ Product View

1 Filter | Date

		Measures	Revenue GC	Sales Quantity	Costs GC
Date	Customer	Product			
2019	(all)	(all)	155,915,470.89 $	71,343 PC	81,910,953.42 $
2018	(all)	(all)	145,269,843.27 $	70,302 PC	81,108,230.65 $
2017	(all)	(all)	152,345,747.48 $	70,599 PC	81,825,384.92 $

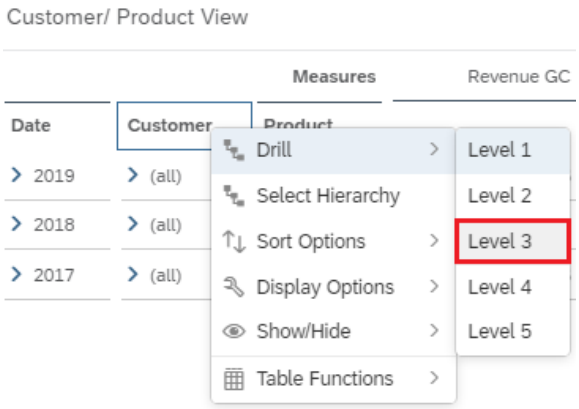
Show/Hide ~~Table Details~~

Next, you remove table details. When you click on your table, the **option More Actions** appears in the upper right corner. Click the button. In the context menu, choose **Show/Hide**. In the following submenu, deselect **Table Details**.



Customer → Drill → Level 3

The customer hierarchy should be open in the initial view up to the level of the sales organization (level 3). To do this, right-click **Customer ► Drill ► Level 3** in the table.

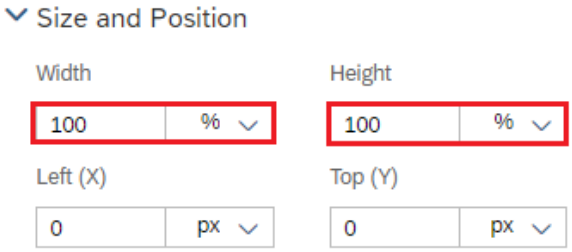


In the *right-hand side panel*, switch from the *Builder* panel to the *Styling* panel. To do so, click **Styling** .

Styling

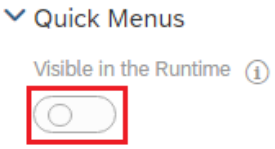
Under *Size and Position*, change *Widht* to **100%** and *Height* to **100%**.

100%

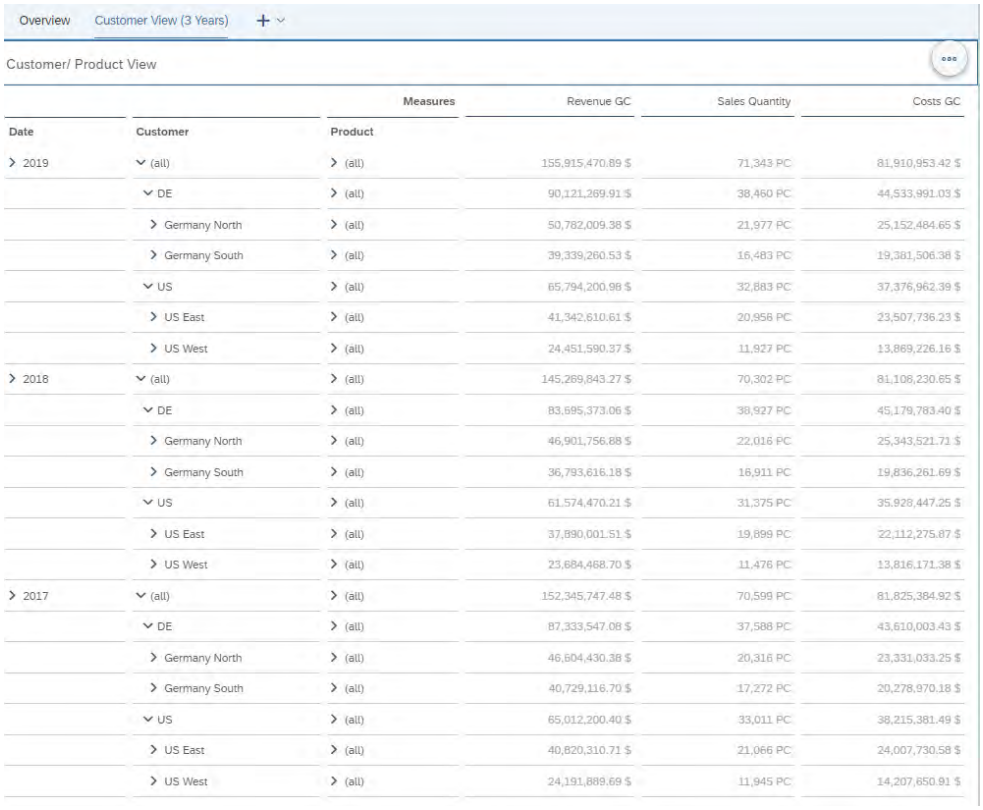


You do not want a context menu to be available in view mode, so you deselect the option under *Quick Menus* in the right-hand *side panel*.

~~Quick Menus~~



With that, you've built a simple, analyzable table on the second page of your story.



Finally, in the menu bar in the *File* area, choose Save icon again and then **Save** to save your changed story.

Save

Step 7: Filter

Task Create different filters. **Time** 15 min.

Description Create a story filter and a new page with Page Filters.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Filter Types

You can use the following three types of filters in an SAC story:

Story Filter

- The *story filter* affects all pages and members of a story, so it should not overlap or exclude from the other two types of filters. An important prerequisite for the end-to-end function of the story filter is that all elements are based on one data source or all data sources for the story filters are linked correctly.

Page Filter

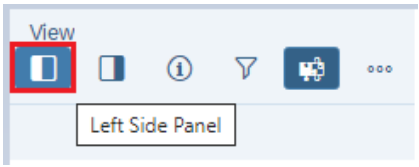
- As the name suggests, the *page filter* works on all elements of a specific page at page level (per tab). Here, too, the general conditions apply for a correct function.

Widget Filter

- The *Filters widget* is the deepest filter level. It is defined at the level of individual elements (widgets). In particular, make sure that there are no collisions with existing story filters and page filters based on the same dimensions or measures.

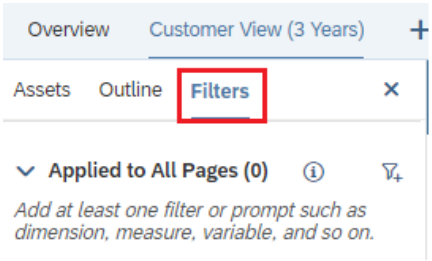
Left Side Panel

You still have the *Customer View (3Years)* page open. In the upper menu area under *View*, open the **Left Side Panel**.



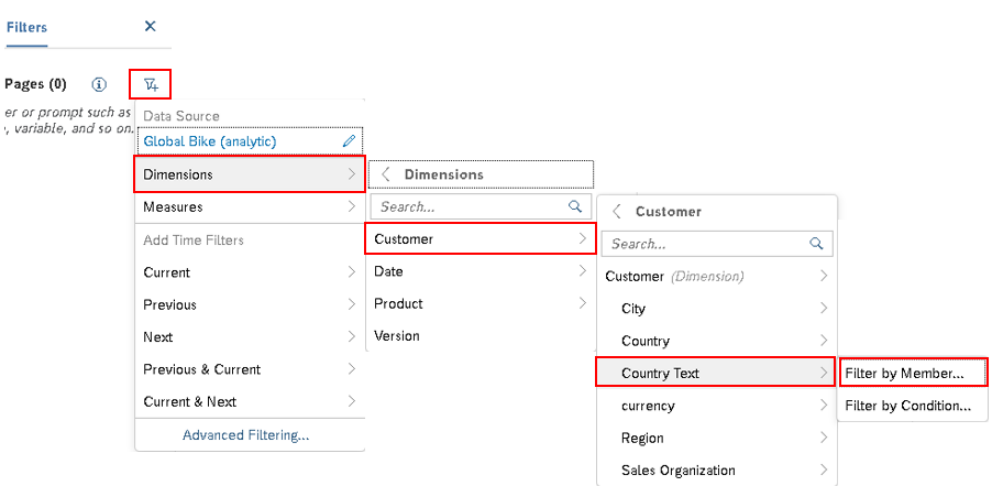
Filters

For the story of this case study, you first define a story filter on the dimension *Country Text*. To do this, switch to the **Filters** tab in the *left-hand side area*.



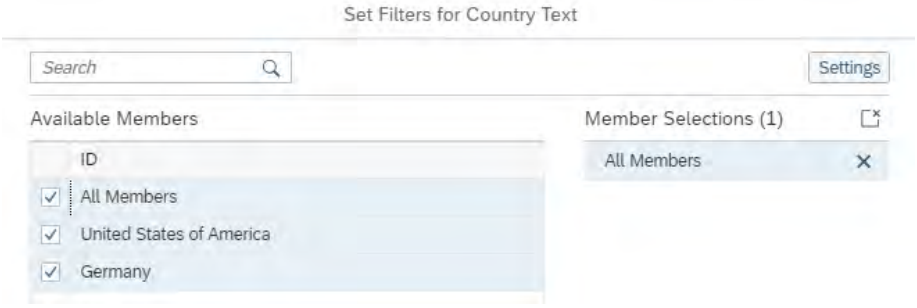
Add New Filters
Dimensions →
Customer→
Country Text →
Filter by Member…

Next *to Applied to All Pages (0)*, click **Add New Filters** ( ) and set a *story filter* to *Country Text* by selecting this attribute via **Dimensions** ▶ **Customer** ▶ **Country Text** ▶ **Filter by Member…**.



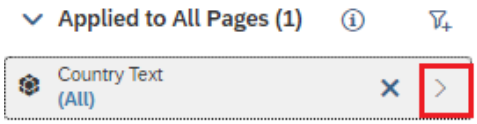
In the dialog box that appears, choose **All Members** and confirm the window by choosing **OK**.

All Members
Ok then



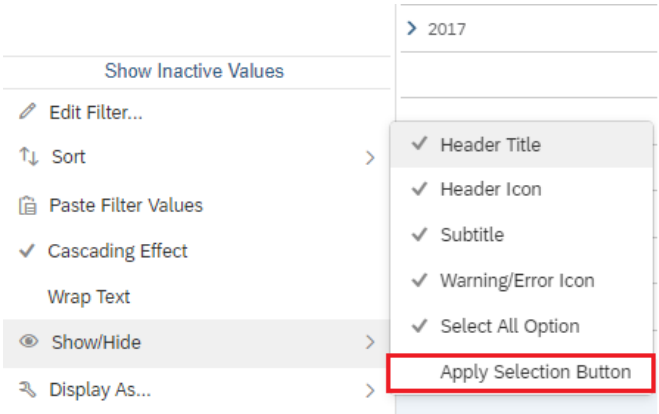
Change the story filter so that the selection is implemented immediately (without confirmation using the *Apply Selections* button). Click **Expand** of the filter.

Expand



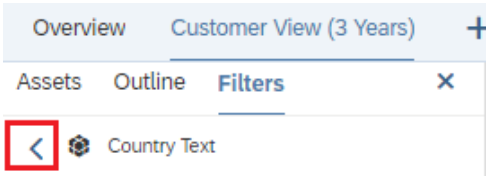
To deactivate the button, select **Show/Hide** ► **Apply Selection Button**.

Show/Hide → ~~Apply Selection Button~~



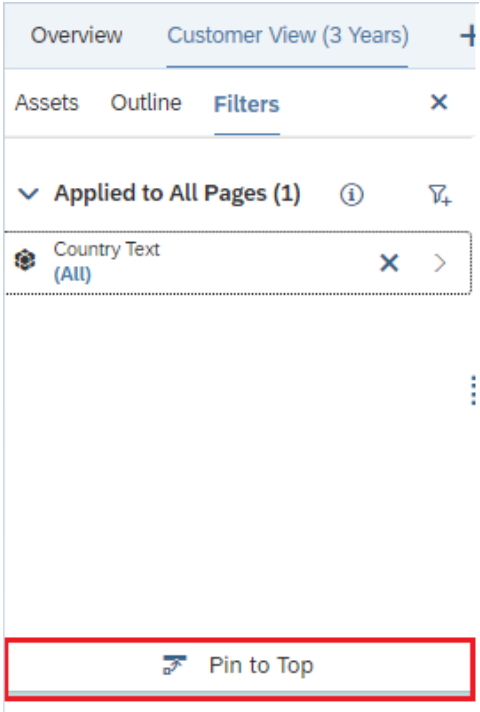
Return to the basic view by choosing **Back** .

Back

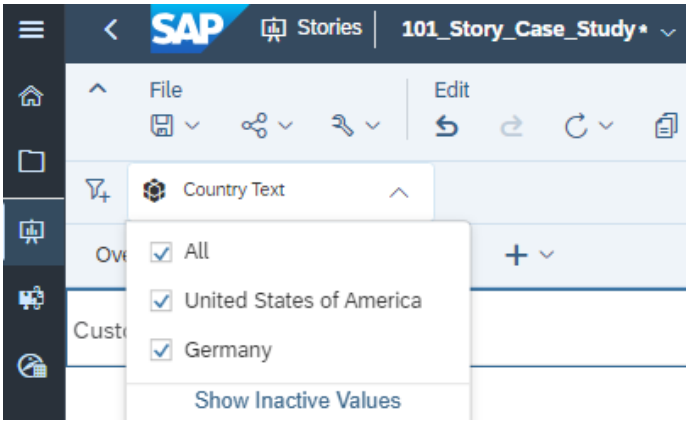


To have the story filter available in the story header, choose **Pin to Top** in the footer of the *Filters* tab.

Pin to Top



In the story header, you have made it possible to filter *text* for all pages using the Country dimension.

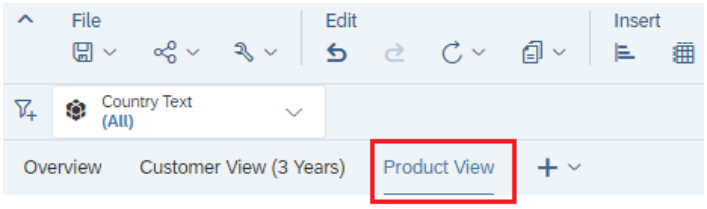


In the *File* area of the ribbon, choose the *Save* icon and then **Save** to save your changed story.

Save

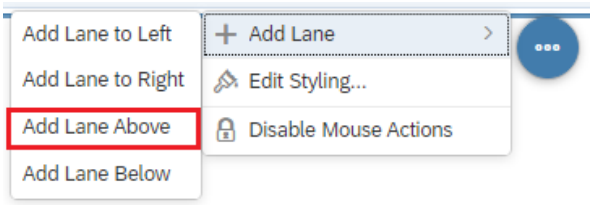
Next, create a new *Responsive* page and name it **Product View**.

Page Product View



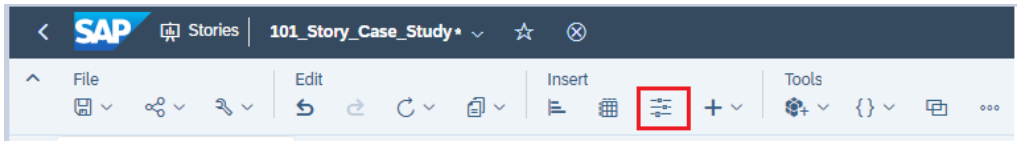
Define page filters for *Date, Sales Organization*, and *Customer*. To do this, you first add a new area above the existing area. To do this, click on the existing area and choose **More Actions** ► **Add Lane** ► **Add Lane Above.**

More Actions → Add Lane → Add Lane Above



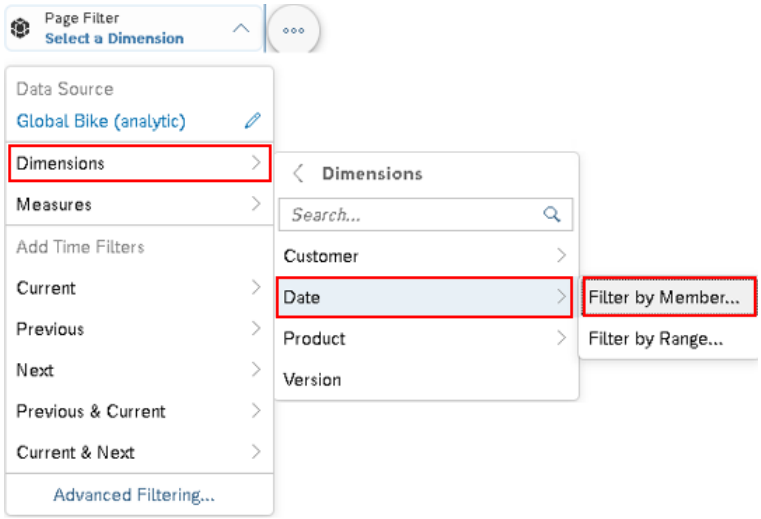
In the new header area of the page, insert the filters. To do this, click in the header area and insert the three filters mentioned above by clicking on the icon **input control** in the *Insert* ribbon.

Input Control



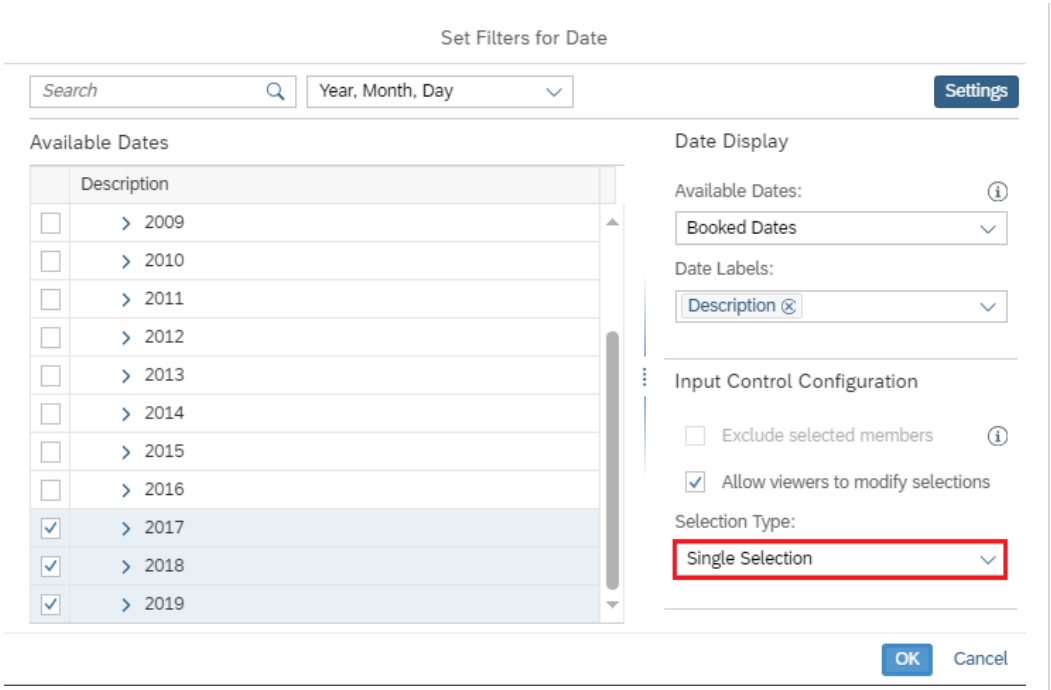
The system then inserts a *Page Filter* element. You must now define this. To do this, select the entry **Dimensions** ► **Date** ► **Filter by Member…** in the menu that opens.

Dimensions → Date → Filter by Member…



In the following dialog, you select the values for **2017, 2018,** and **2019** and the *selection type* as **single selection,** since only a single selection of the year should be possible in this filter. Confirm the window with **OK.**

2017, 2018, 2019
Settings
Single Selection
Ok then



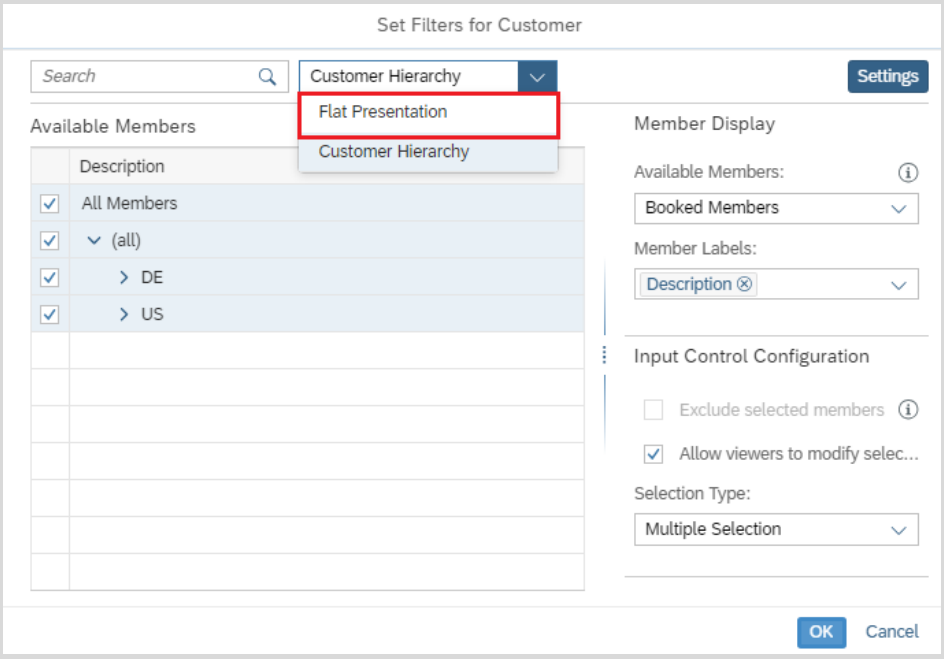
Repeat the steps for **Sales Organization** (under *Customer) Filter by Member…* with selection *All Members* with option *Multiple Selection*.

Sales Organization

Multiple Selection specifies that the end user should be able to select multiple options at the same time.

For **Customer (Dimension)**, also proceed as a *filter by member….* Deactivate the hierarchy with **the Flat Presentation selection**, choose *All Members* with the *Multiple Selection* **option**.

Flat Presentation



To save space, you want to hide the unnecessary icons next to the filter titles. To do this, right-click the relevant filter and choose **Show/Hide ► Header Icon**.

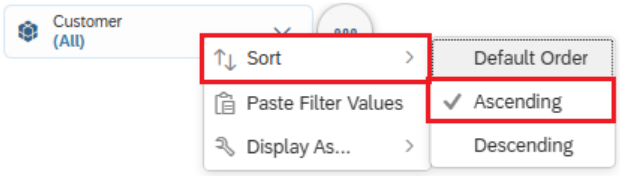
Show/Hide → Header Icon

For the filters on *Sales Organization* and *Customer*, disable the *Apply Selection* button by right-clicking in the filter and selecting **Show/Hide ► Apply Selection Button**.

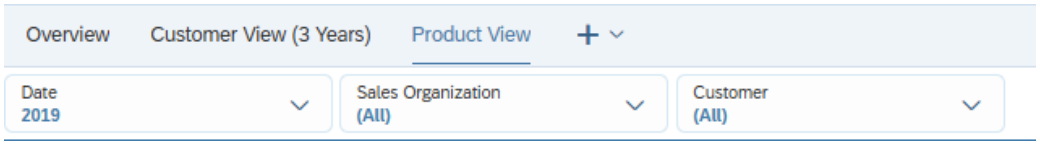
Show/Hide → ~~Apply Selection Button~~

Right-click the filter *Customer* and choose **Sort ► Ascending** to sort the customers alphabetically.

Sort → Ascending



As a result, you can now find the three defined page filters in your page header.



In the lower area, you now want to insert a table that shows you the *date* and *product* information.

Click in the area below the filters and choose the Icon **Table** in the *Insert* area of the ribbon.



The *right-hand side panel* in the *Builder* panel has now opened.

In the *Rows* section, choose **Add Dimensions** and, one by one, the **Date** and **Product** dimensions.

Add Dimensions
Date
Product

You now define the key figures to be displayed in your table. To do so, move your mouse to the *Columns* section and click **Manage Filters** next to *Measures*.

Manage Filters

In the window that opens, select the key figures **Sales Quantity** and **Contribution Margin GC**, and confirm the window by choosing **OK**.

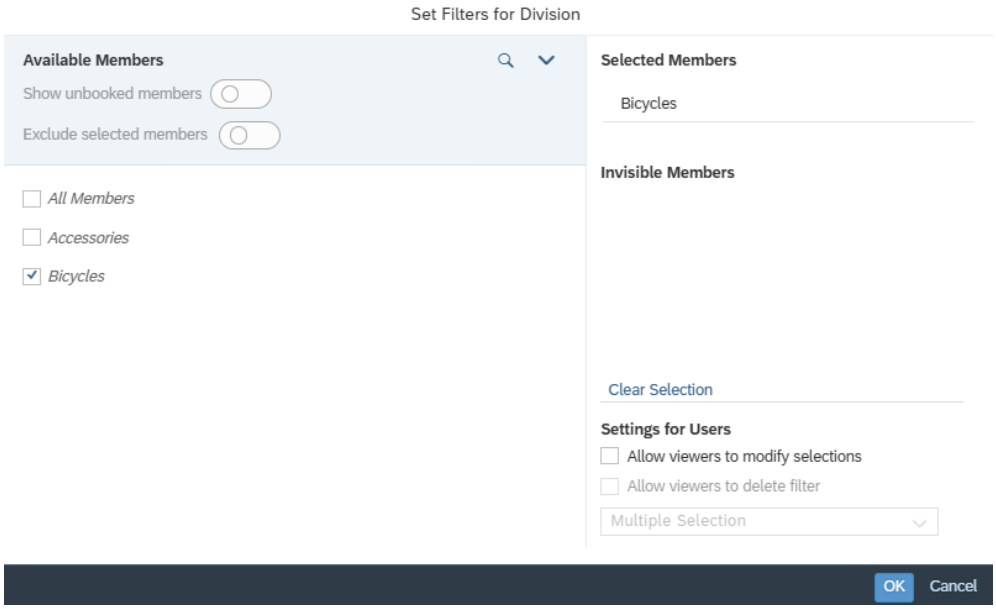
Sales Quantity
Contribution Margin GC

Since the focus of this page is on the *Bicycles division*, choose **Add Filters** in the *Filter* area. Now choose **Dimension ► Product ► Division.** In the *Set Filters for Division* window, select the **Bicycles** *division*.

Add Filters
Product →
Division ->
Bicycles

Since this should be a rigid filter, that is, a fixed filter that cannot be changed by the end users, deselect **Allow viewers to modify selections** and confirm the window with **OK**.

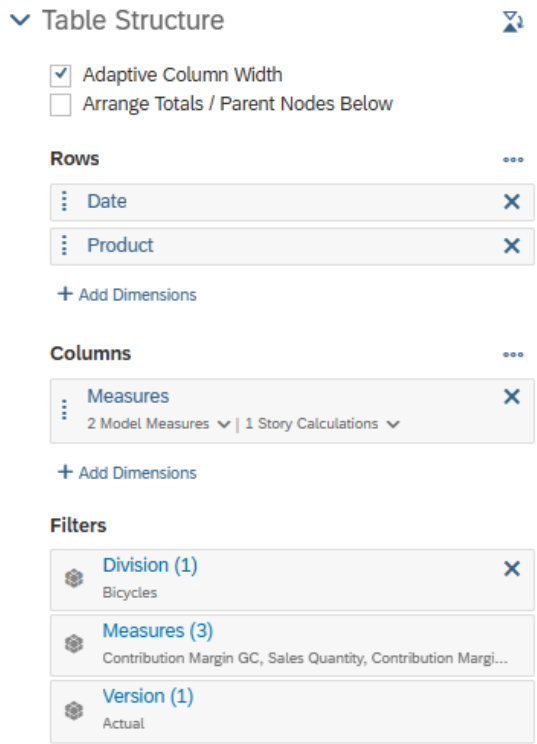
~~Allow viewers to modify selections~~



Save

In the *File* area of the ribbon, choose *Save* icon and then **Save** to save your changed story.

Compare your settings with the following screen.



Bicycles

Now adjust the header of the table. To do this, double-click in the text and enter **Bicycles** instead of the model name.

Next, you remove table details. When you click on your table, the **option More Actions** appears in the upper right corner. Click the button. In the context menu, choose **Show/Hide**. In the following submenu, deselect **Table Details**.

More Actions
Show/Hide
~~Table Details~~

In the table, click the arrow below *Date* to expand its data to the level of the year.

In the *right-hand side panel*, switch from the *Builder* panel to the *Styling* panel. To do so, click **Styling**.

Styling

In view mode, the users should not have a context menu available.

In the *Quick Menus* area on the *right-hand side* of the screen, deselect it.

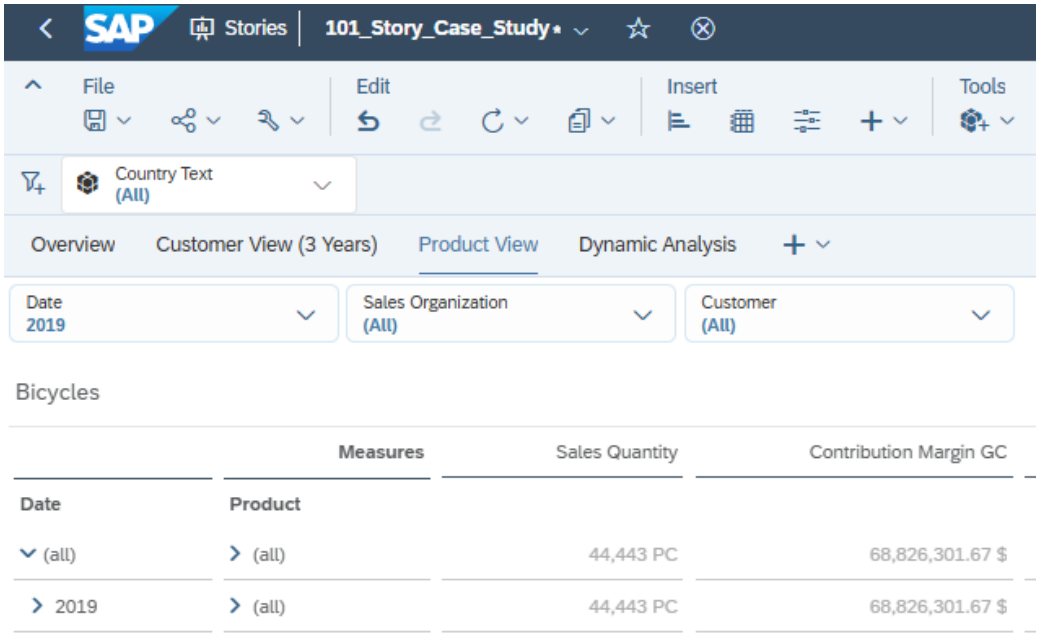
~~Quick Menus~~

Also deselect the *Quick Menus* for all 3 page filters.

~~Quick Menus~~

Test the functionality of your page filter by selecting the year **2019** for *Date*.

2019



Finally, choose *Save* icon again in the *File* area of the ribbon and then **Save** to save your changed story.

Save

Step 8: Graphics (Heat Map)

Task Create a Heat Map. **Time** 10 min.

Description On the *Product View* tab, create a heat *map* graphic.

You are still in the *story Story Case Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Initial Screen

On the *Product View* page, you first insert a new area next to the area of the table. To do this, click the area of the table and choose **More Actions** ► **Add Lane** ► **Add Lane to Right.**

More Actions →
Add Lane →
Add Lane to Right

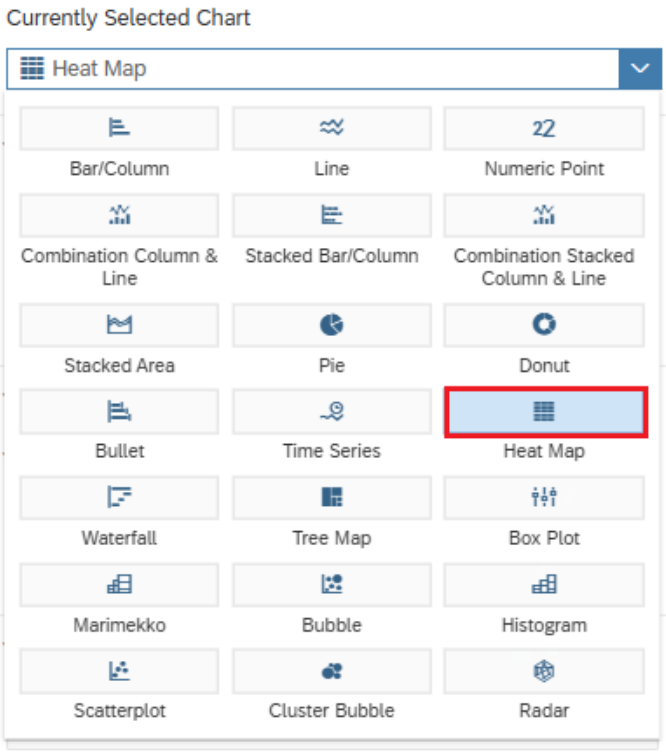
In the new area, use the ribbon to insert a graphic in the *Insert* area by clicking the **Chart** icon .

Insert -> Chart

The *right-hand side panel* in the *Builder* panel has now opened.

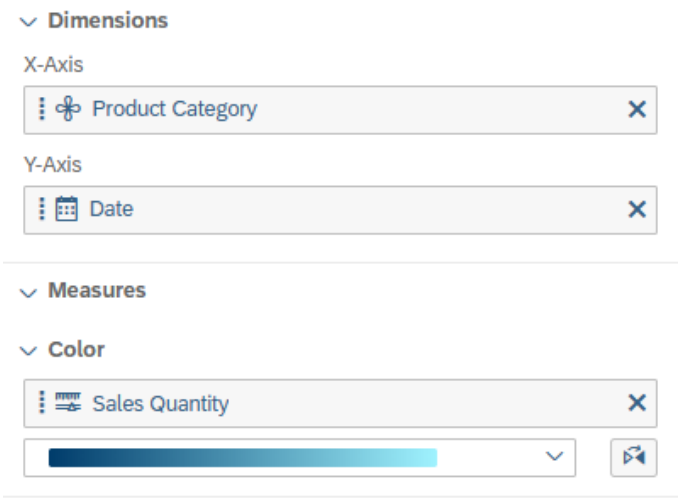
Change the graphic type to *Heat Map*. To do so, open the dropdown under **Current Selected Chart** and select the **Heat Map** entry.

Current Selected Chart
Heat Map



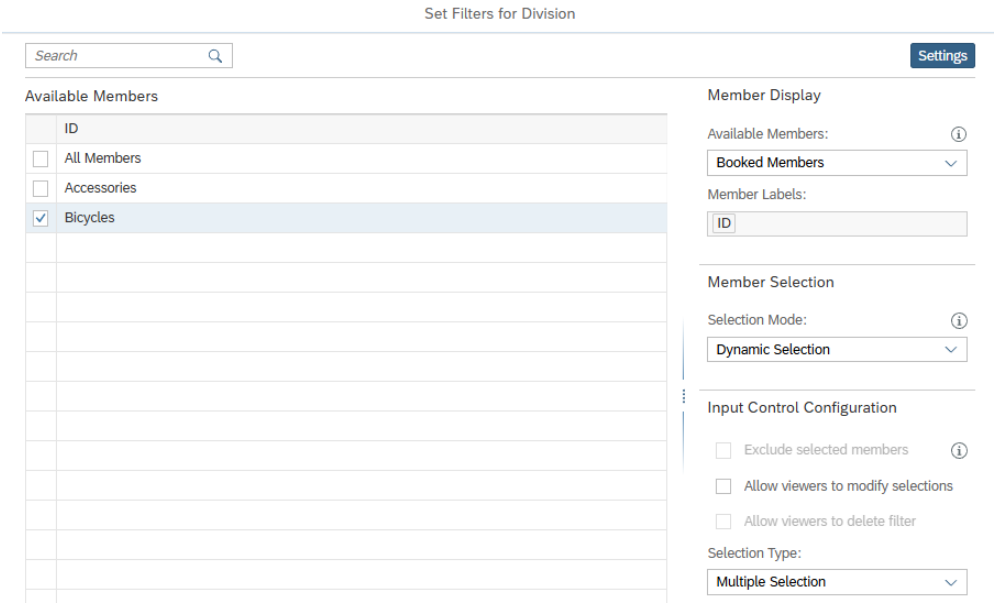
Define **Product Category** (under *Product*) as *X-Axis*, **Date** as *Y-Axis*, and **Sales Quantity** as *Color*.

Product Category
Date
Sales Quantity



Under *Add Filters* for the Division dimension, add a filter on **bicycles** that cannot be changed by the end users.

Add Filters
Bicycles

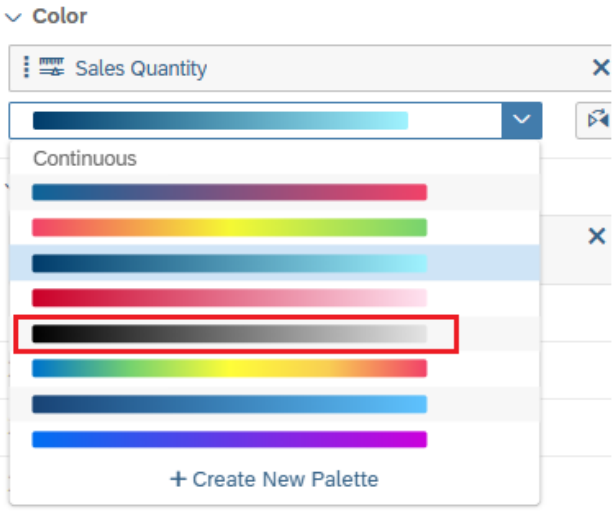


Since you want the dimension *Date* to be displayed after months, now use the mouse to move to the area of the Dimension *Date* and select **Hierarchy** and then **Level 3**.

Hierarchy
Level 3

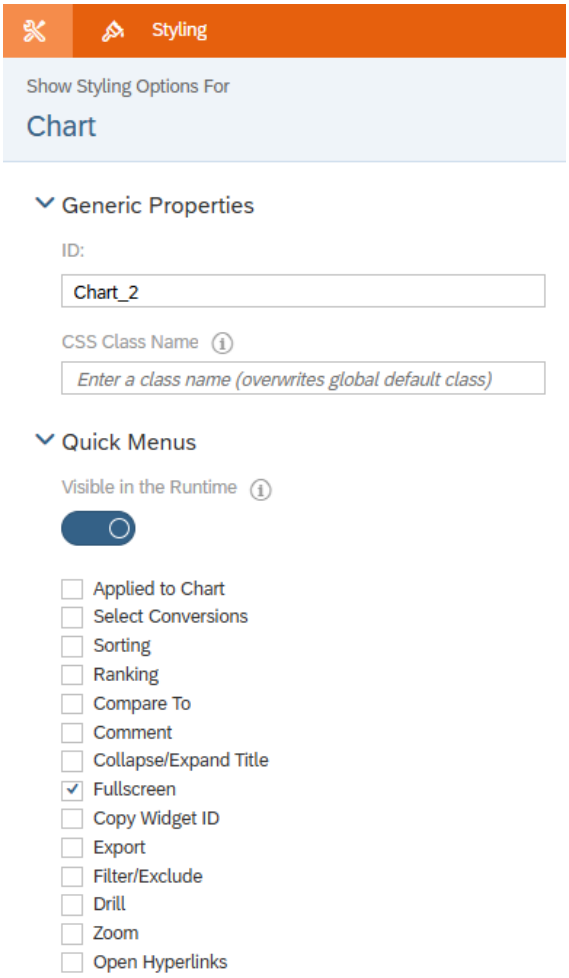
Next, you change the *color gradient* to **grayscale** using the dropdown.

Grayscale



In the *right-hand side panel*, switch from the *Builder* panel to the **Styling** panel, and in the *Quick Menus* area, deselect all options except **Fullscreen**.

Styling
Fullscreen



Now adjust the heat *map* header. To do this, double-click in the text and enter **Quantity for Bicycles**.

Quantity for Bicycles

Quantity for Bicycles

in PC

	E Bikes	Offroad Bikes	Road Bikes	Touring Bikes	Trend Bikes
Jan	39	359	340	584	5
Feb	67	837	603	761	20
Mar	69	696	581	750	10
Apr	178	1,874	1,715	1,831	67
May	330	2,213	1,874	2,183	73
Jun	368	3,170	2,485	3,364	85
Jul	164	1,457	1,193	1,656	5
Aug	127	1,469	1,212	1,401	61
Sep	149	1,097	937	1,467	44
Oct	83	737	685	785	20
Nov	30	409	310	310	17
Dec	65	410	295	273	44

Finally, in the menu bar in the *File* area, choose *Save*  icon again and then **Save** to save your changed story.

Save

Step 9: Graphics (Donut)

Task Create so-called donut graphics. **Time** 10 min.

Description Create a graphic of type *Donut* on the *Product View* tab.

Initial Screen

You have still opened the *Story_Case_Study* story from the previous step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

More Actions → Add Lane → Add Lane Below

On the *Product View* page, you first insert a new area below the area of the page filters. To do this, click the area of the page filters and choose **More Actions** ► **Add Lane** ► **Add Lane Below.**

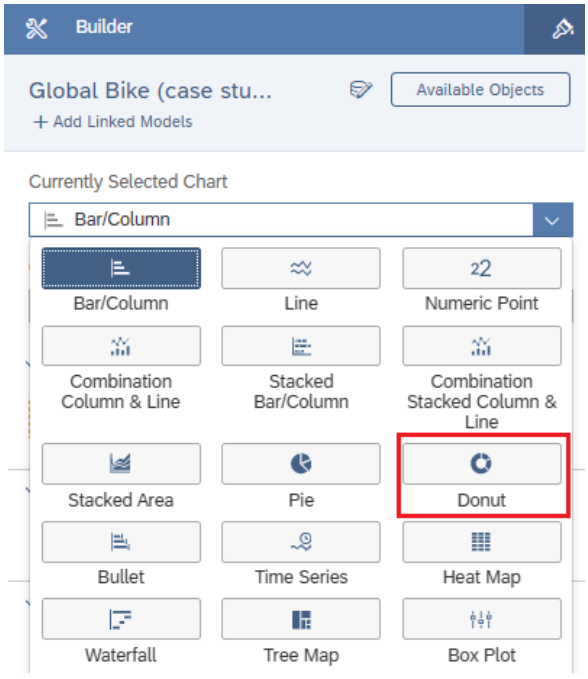
Chart

Insert a graphic in the new area by clicking the **Chart** icon ( ) in the *Insert* area of the ribbon.

The *right-hand side panel* in the *Builder* panel has now opened.

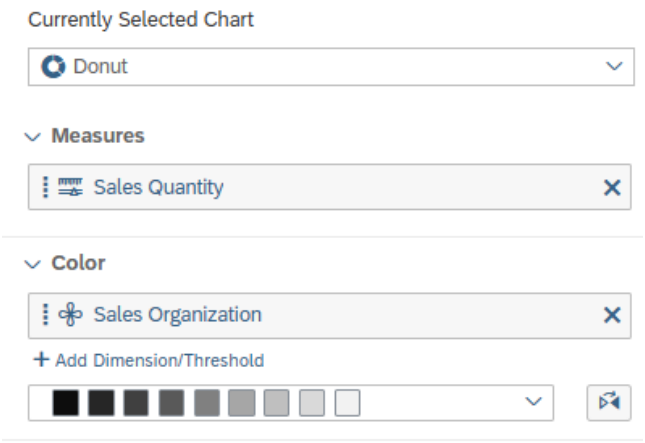
Current Selected Chart Donut

Change the graphic type to *Donut*. To do this, open the dropdown under **Current Selected Chart** and choose **donut**.



Sales Quantity
Sales Organization
Grayscale

You now define **Sales Quantity** as *Measure*, **Sales Organization** as *Color*, and **Grayscale** as *Color Scheme.*

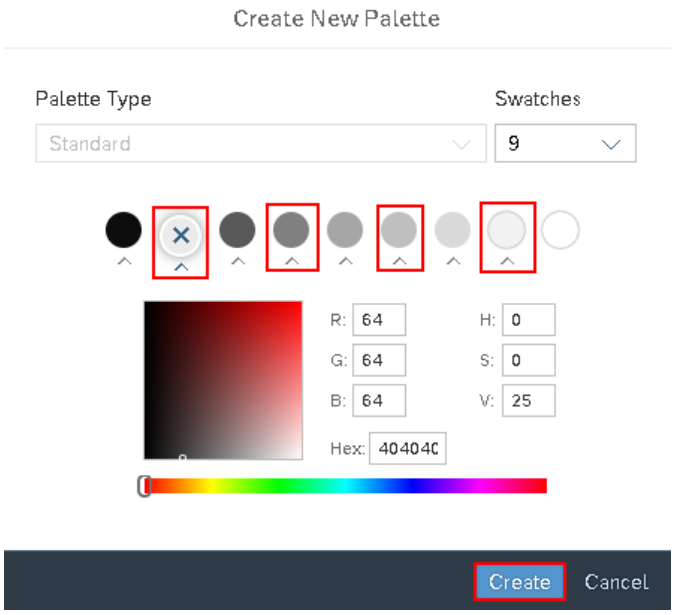


Since the different shades of gray are difficult to distinguish, change the color palette. To do this, choose + next to the color palette.

+

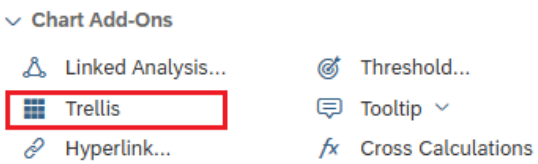


The *Create New Palette* window opens and now delete every 2. Color by clicking on it and choosing **x**. Finally, close the window by choosing **Create**.



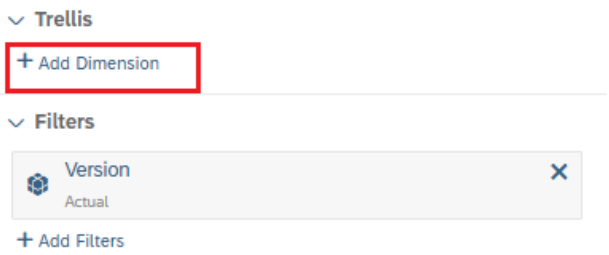
To display a graphic for each *product category*, use the *Trellis* option. Open the Chart Add-Ons area and then choose Trellis.

Chart Add-Ons
Trellis



The *Trellis* area has now been added. Click **Add Dimension** and select **Product Category** (under *Product*).

Add Dimension
Product Category



Sales Quantity per Category (3 Years)

On the *right-hand side panel*, switch from the *Builder* panel to the **Styling** panel and deselect *Quick Menus*.

Styling
~~Quick Menus~~

Under *Size and Position*, change *Widht* to **36 Grid** and *Height* to **11 Grid**.

36 Grid
11 Grid

In the *right side panel*, scroll down to the *Legend* pane. Under *Placement*, choose **Beside Chart (Right)**.

Beside Chart (Right)



Next, disable the display of all *subtitles*. When you click on your graphic, the **option More Actions** ( ) appears in the upper right corner. Click the button. In the context menu, choose **More Options ► Show/Hide ►Subtitle ► All.**

More Actions
More Options →
Show/Hide →
Subtitle →
~~All~~

Change the title of the graphic to **Quantity per Sales Organization and Category**.

Quantity per Sales Organization and Category



Finally, in the menu bar in the *File* area, choose *Save* icon again and then **Save** to save your changed story.

Save

Step 10: Dynamic Graphics

Task Create a dynamic graphic. **Time** 15 min.

Description Create a dynamic graphic in which the end user can select the key figures and characteristics of the analysis view at the time of execution.

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Initial Screen

So that the end user can dynamically analyze their measure and dimension on a time axis of years, you want to make a dynamic evaluation option available in your story next.

To do this, you define the *Dynamic Analysis* page by creating a new *Responsive* page and renaming it to **Dynamic Analysis**.

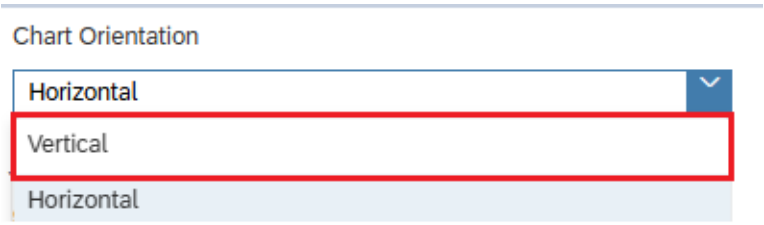
Add New Page
Dynamic Analysis

Use the ribbon to insert a graphic in the *Insert* area by clicking the **Chart** icon.

Chart

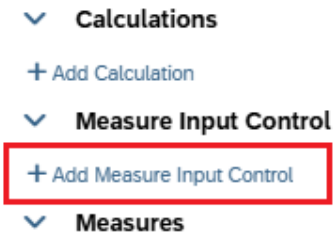
The *right side panel* in the *Builder* panel opens. Leave the graphic type and change the orientation to **Vertical** using the **Chart Orientation** dropdown.

Chart Orientation
Vertical



Define dynamic key figure selection. To do so, select **At least 1 Measure required** under *Measures* in the *Builder* panel on the *right side panel*, and then select **Add Measure Input Control**.

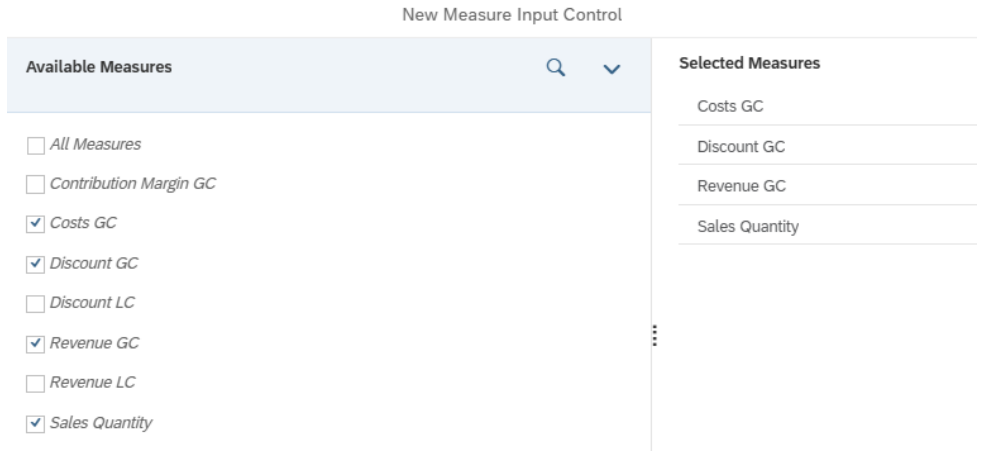
At least 1 measure required
Add Measure Input Control



In the window that appears, select the key figures **Costs GC**, **Discount GC**, **Revenue GC**, and **Sales Quantity** and confirm the window by choosing **OK**.

Costs GC
Discount GC
Revenue GC

Sales Quantity
Ok then



Add Lane to Left

A selection filter now appears to the right of your graphic. Move it to the left side of the graphic. Insert a new area to the left of the graphic and move the selection filter to it.

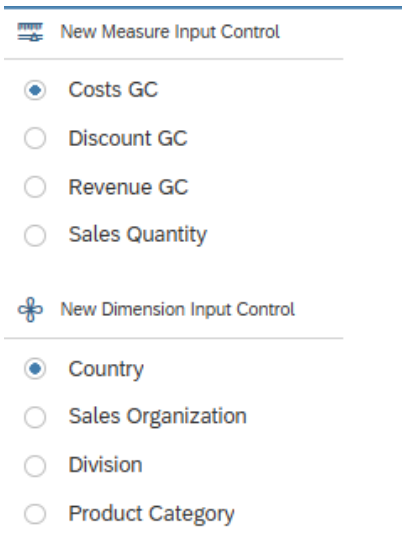
Add Dimension
Add Dimension Input Control

To change the dimensions, click on the graphic again and in the *Builder*, under *Dimensions*, choose **Add Dimension** and then choose **Add Dimension Input Control**.

Country
Sales Organization
Division
Product Category
Ok then

In the window that appears, select the dimensions **Country**, **Sales Organization** (both under *Customer*) and **Division** and **Product Category** (both under *Product*) and confirm the window with **OK**.

You also move this selection filter to the left, below the selection filter of the key figure, with a little distance. Next, drag both selection filters so that the selection can be made easily via radio buttons.



Define a *Page Filter* on *Date*, which enables end users to select a maximum of the last five years.

Input Control

To do this, click in the right-hand area and insert a filter using the *Insert* ribbon and click on the icon **input control** .

The Page *Filter* element is then inserted. Define the Page Filter. To do this, select the entry **Dimensions ► Date ► Filter by Member…** in the menu that opens.

Dimensions →
Date →
Filter by Member…

To do this, click in the header area and insert a filter using the menu **Insert ► Input Control**.

Insert -> Input Control

This inserts a *Page Filter element*. You now need to define this. To do this, in the menu that opens, select **Dimensions ► Time ► Filter by Members…**.

Dimensions -> Time ->
Filter by Member…

In the following dialog, select the **last five years (2015-2019)**, select **Multiple Selection** as the Selection Type under Settings, and confirm the window with **OK**.

last five years
Multiple Selection
Ok then

Now drag the filter below your dimension filter on the left and adjust it to the full size as well.

For the filter on *Date*, deactivate the *Apply Selection* button and the *Cascading Effect* by right-clicking in the filter and **choosing Cascading Effect or Cascading Effect**. **Show/Hide ► Apply Selection Button**.

~~Cascading Effect~~
Show/Hide →
~~Apply Selection Button~~

Insert **Date** as the second dimension of the graphic using **Add Dimension**.

Add Dimension
Date

Since you want the Dimension *Date* to be displayed after years, now use the mouse to move to the area of the Dimension *Date* and select **Hierarchy** and then **Level 2**.

Hierarchy
Level 2

Change the color palette to **grayscale**.

Grayscale

In the *right-hand side panel*, switch from the *Builder* panel to the **Styling** panel and resize the *Date Page Filter* by clicking it and changing the *Height* to **13 Grid** under *Size and Position*.

Styling
Date Height 13 Grid

Under *Quick Menus*, select **Visible in the Runtime** for all 3 filter elements.

~~Quick Menus~~

Rename the selection filter for the measure in the **measure** and the selection filter for the characteristic to **Second Dimension**. To do this, double-click the relevant text. Once it's marked in blue, you can adjust the text.

Measure
Second Dimension

Adjust the width of the left panel to the width of the filters.

Change the size of the diagram by clicking it and changing the *Width* to **25 Grid** and the *Height* to **30 Grid** under *Size and Position*.

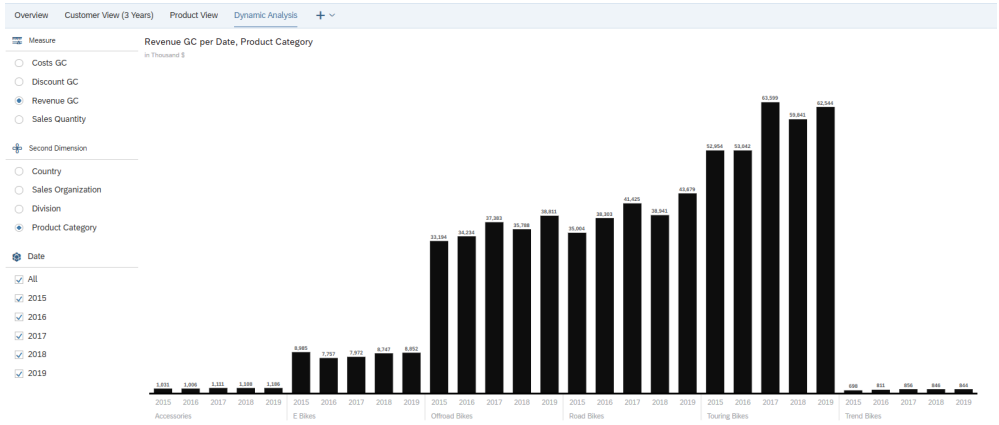
Width 25 Grid
Height 30 Grid

Deactivate the *quick menus* for the graphic.

~~Quick Menus~~

To keep the key figures readable, you need to adjust their scaling. To do this, scroll down to the *Number Format* area and select **Thousand** for *Scale* and **0** for *Decimal Places*.

Thousand
0



Finally, in the menu bar in the *File* area, choose Save  icon again and then **Save** to save your changed story.

Save

Step 11: Calculated Key Figures

Task Create calculated key figures. **Time** 10 min.

Create the calculated key figure *Contribution Margin %* in the table on the *Product View* tab page.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Calculated Key Figures

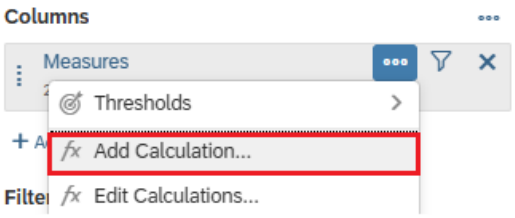
SAC allows you to create *calculated measures* for a model within a story. For these calculations, you can use all measures in the model as operands.

For a better analysis option of the product data, the key figure *Contribution Margin %* is to be added as a calculated key figure in the table on the *Product View* tab page.

To create the calculated contribution *margin %* key figure, go to the table on the *Product View* tab and open the *Builder* panel on the *right-hand side panel*.

Add Calculation…

Now move your mouse to the *Columns* section and then click *More* ••• next to *Measures*. Now choose the **Add Calculation…** entry.



Calculated Measure Contribution Margin %

The *Calculation Editor window opens*. As *the type*, first select **Calculated Measure**. Enter **Contribution Margin %** as the *name* of the calculated key figure.

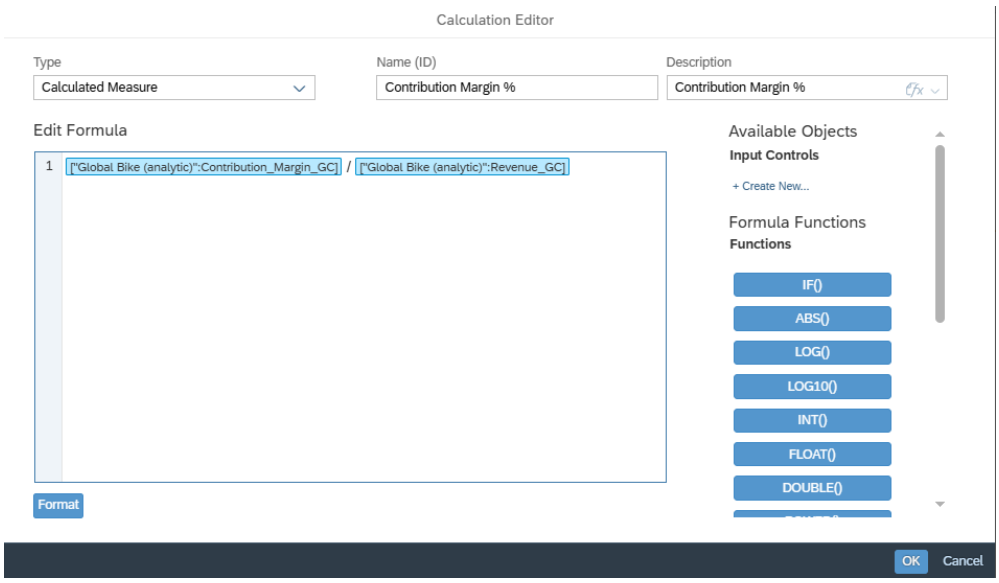
Formula

In the formula field (*Edit Formula* section), you define the formula **Contribution Margin GC/ Revenue GC**.

Note To insert the key figures, use the key combination *CTRL + Space* as input help. The measures included in the model are at the bottom of the list.

Ok then

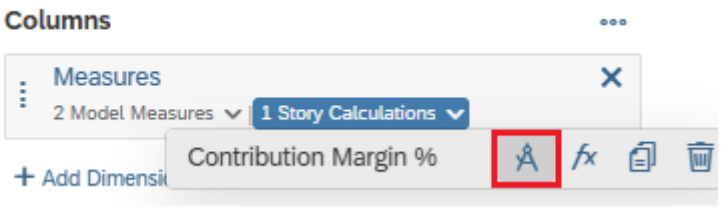
When you have finished defining your measure, confirm the window with **OK** so that the new measure is applied to your story.



Your new *Contribution Margin %* key figure is immediately inserted into your table.

After saving the calculated *Contribution Margin %* key figure, you must adjust the number formatting to display the key figure as a % value.

Move your mouse to the *Columns* section, then follow the path *1 Story Calculations* ► *Contribution Margin %* and click **Edit formatting options** .

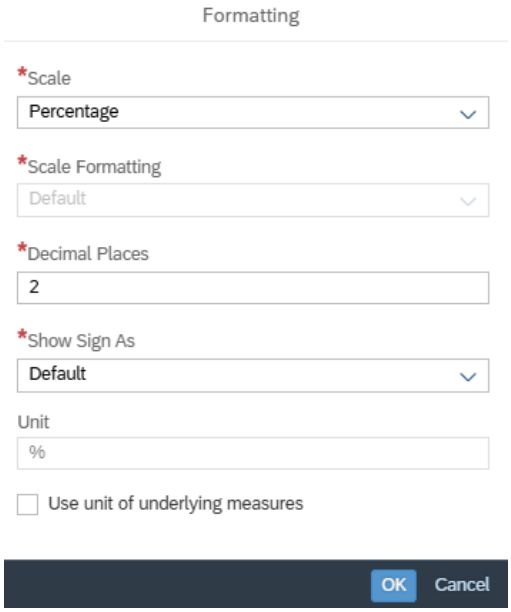


In general, the number format of the underlying measures in the model is always applied. You must deactivate this behavior in the *Formatting* window by deselecting the option **Use unit of underlying measures**.

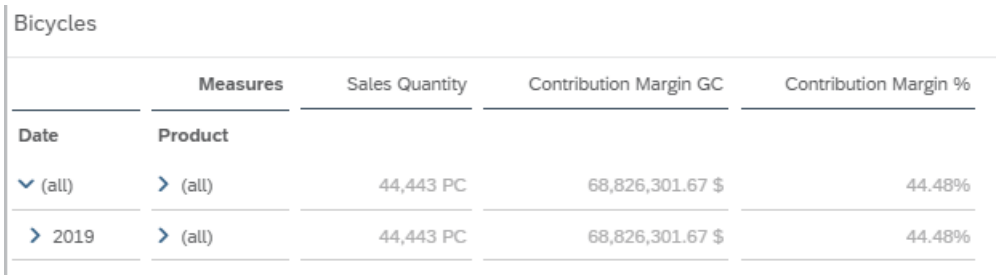
~~Use Unit of Underlying Measures~~

You can then select a different number format (in our case**, Percentage**) in the *Scale* menu option. Confirm the adjustments with **OK**.

Percentage
OK



You now increase the size of the table to the full width of the area so that all key figures can be read at first glance.



Finally, in the menu bar in the *File* area, choose Save icon again and then **Save** to save your changed story.

Save

Step 12: Restricted Key Figures

Task Create a restricted key figure **Time** 10 min.

Description Create the restricted key figure *Revenue Accessories* and use it in the calculated key figure *% Revenue Accessories* in the table of the *Customer View (3 Years)* tab.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Restricted Measures

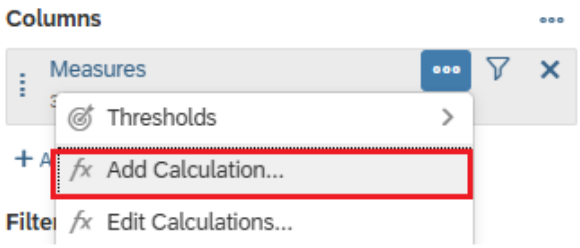
In SAC, in addition to calculated measures, you also have the option to create *restricted measures* for a model within a story. All key figures and characteristics (dimensions) defined in the model are available for the restrictions.

For a better analysis of the customer buying pattern, the key figure *% Rev Access* (% Revenues Accessories) is to be added to the table of the tab customer *view (3 years*).

The first step is to create the restricted key figure *Revenue Accessories*. To do this, go to the table on the *Customer View (3 Years)* tab and open the right-hand side panel and the *Builder* panel.

More
Add Calculation…

Now move your mouse to the *Columns* section and click **More** next to *Measures*. Now choose the **Add Calculation…** entry.

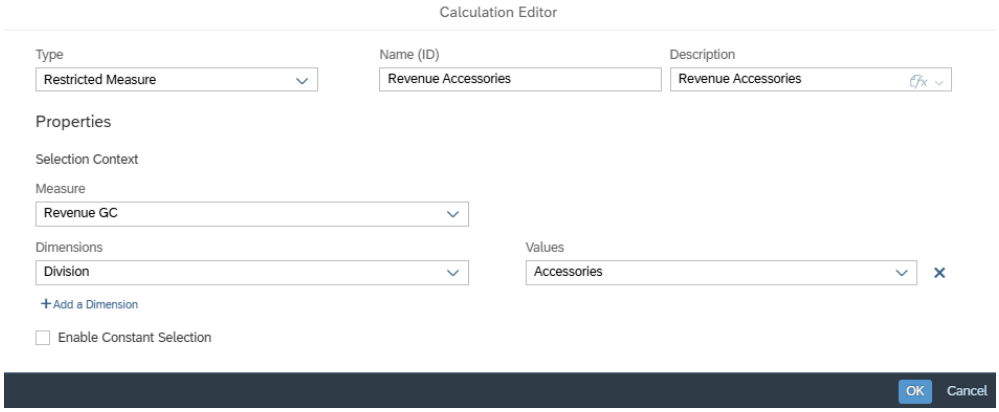


Restricted Measure
Revenue Accessories

In the *calculation editor* that opens, select **Restricted Measure** as the *type*. Enter **Revenue Accessories** as the *name*.

The underlying logic is:

- Key Figure: *Revenue GC*
- Dimension: *Division* (under *Product*)
- Value: *Accessories (Select by Member…)*

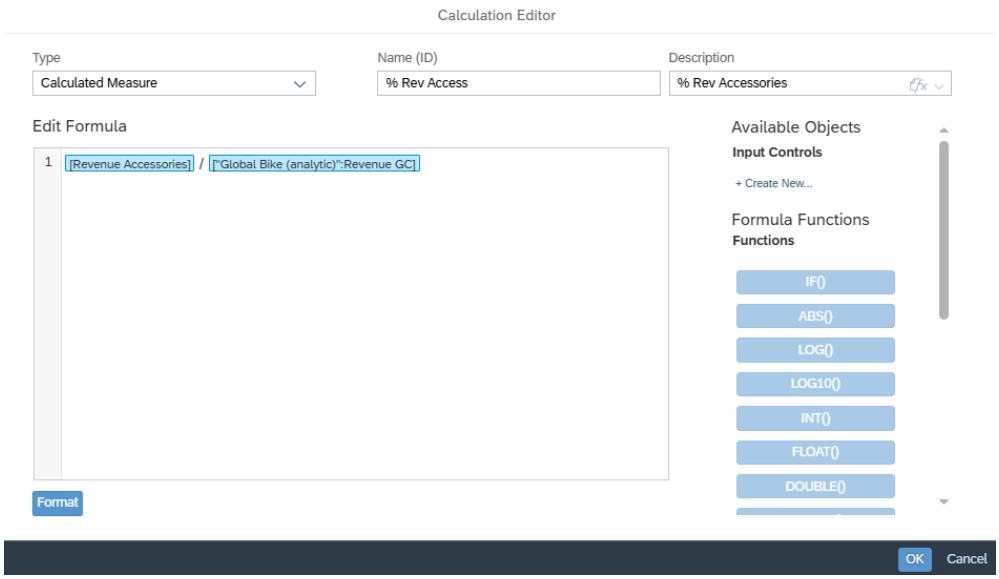


When you have completely defined your key figure, confirm the window with **OK**.

Ok then

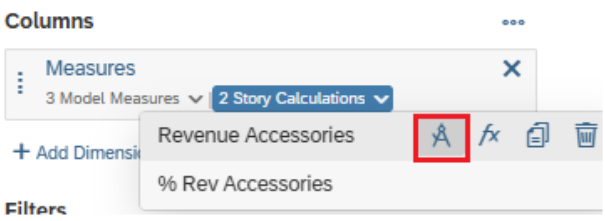
Next, you create the calculated key figure *% Rev Access*. Proceed as for the *Contribution Margin %* key figure. The underlying formula is **Revenue Accessories / Revenue GC**.

Formula



For the % *Rev Access* key figure, you must now set the output as a % value. To do so, move your mouse to the Columns section, follow the path *2 Story Calculations* ► *% Rev Access*, and click **Edit formatting options** .

1 Story Calculations → % Rev Access -> Edit Formatting Options



In the *Formatting* window, deselect the entry **Use unit of underlying measures** to be able to use a different format for the key figure and select the entry **Percentage** for *Scale*. Finally, confirm the window with **OK**.

~~Use Unit of Underlying Measures~~
Percentage
Ok then

Since only the calculated measure is to be displayed in the table, hover over the *Columns* section and click Manage **Filters** next to *Measures*.

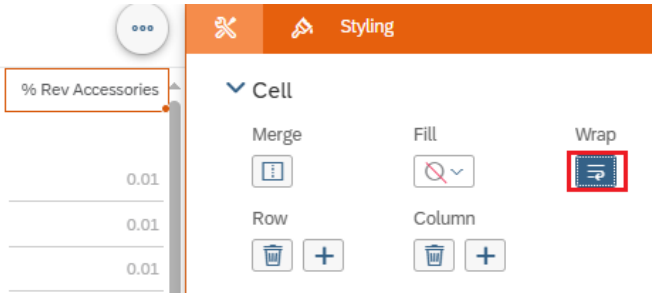
Manage Filters

Deselect the **Revenue Accessories** entry and confirm the window with **OK**.

~~Revenue Accessories~~
Ok then

To prevent the *% Rev Access* column from taking up as much space, set a line break. To do so, click the *% Rev Access* text and open the **Styling** panel in the right-hand side panel. Scroll to *Cell* and select **Wrap**.

Styling
Wrap



You can now reduce the column width according to the width of the key figure values. Now change the width of the table in such a way that all key figures in the table are easily readable at first glance.

Customer/ Product View

		Measures	Revenue GC	Sales Quantity	Costs GC	% Rev Accessories
Date	Customer	Product				
> 2019	v (all)	> (all)	155,915,470.89 $	71,343 PC	81,910,953.42 $	0.01
	v DE	> (all)	90,121,269.91 $	38,460 PC	44,533,991.03 $	0.01
	> Germany North	> (all)	50,782,009.38 $	21,977 PC	25,152,484.65 $	0.01
	> Germany South	> (all)	39,339,260.53 $	16,483 PC	19,381,506.38 $	0.01
	> US	> (all)	65,794,200.98 $	32,883 PC	37,376,962.39 $	0.01
> 2018	v (all)	> (all)	145,269,843.27 $	70,302 PC	81,108,230.65 $	0.01

Finally, in the menu bar in the *File* area, choose Save icon again and then **Save** to save your changed story.

Save

Step 13: Thresholds

Task Create a threshold. **Time** 10 min.

Description Create a threshold for the *Contribution Margin %* measure on the *Product View* tab.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

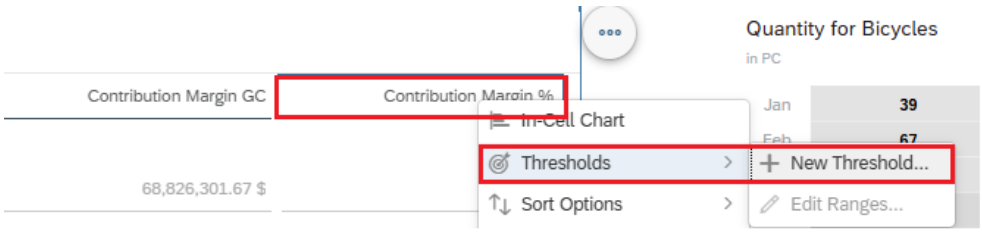
Thresholds

Threshold values are used for the fast optical detection of deviations or for exceeding/falling below defined thresholds. Traditionally, a distinction is made between red (bad) and green (good).

Since the company wants to achieve a contribution margin ratio of 46% on average with its products, add a corresponding threshold value for the *Contribution Margin %* key figure on the *Product View* tab in the table.

Thresholds -> New Threshold…

Switch to the *Product View* tab and click the *Contribution Margin %* heading in the table. Clicking the *right mouse button* opens the menu. Now choose **Thresholds ► New Threshold…**.

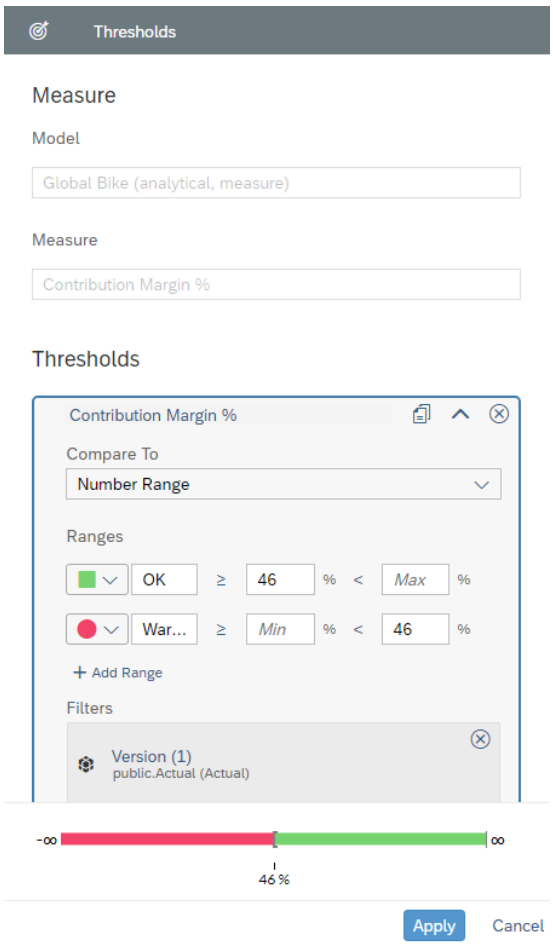


Contribution Margin %

The definition view now opens in the right-hand pane. First, double-click in the text *New Threshold*. If it is marked in blue, you can change it to **Contribution Margin %**.

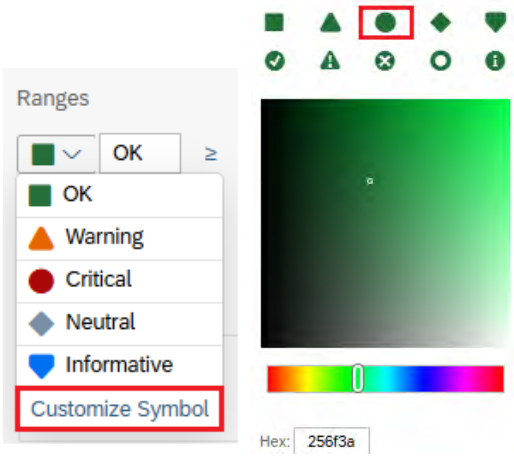
>= 46
Add Range
< 46

Now you define the thresholds. Green is **greater than or equal to 46**. To be able to define the threshold for red, choose *Add Range* and select the *red circle* for the color. You then define the threshold value as **less than 46**.



Since a continuous display is desired, you now change the color green to a circle. To do this, click on the *green square* and choose **Customize Symbol**. Now select the **green circle** in the header area of the window and click in the area outside the symbol selection.

Customize Symbol
Green Circle



To finish defining the threshold, click **Apply** in the lower left corner and close the right side panel.

Apply

Open the hierarchy of the *Product* dimension at product *category* level to check whether your threshold works correctly.

Bicycles

Date	Product	Measures	Sales Quantity	Contribution Margin GC	Contribution Margin %
(all)	(all)		44,443 PC	68,826,301.67 $	44.48%
2019	(all)		44,443 PC	68,826,301.67 $	44.48%
	Bicycles		44,443 PC	68,826,301.67 $	44.48%
	E Bikes		1,669 PC	3,475,672.78 $	39.27%
	Offroad Bikes		14,728 PC	18,276,925.24 $	47.09%
	Road Bikes		12,230 PC	19,885,494.09 $	45.53%
	Touring Bikes		15,365 PC	26,843,992.07 $	42.92%
	Trend Bikes		451 PC	344,217.49 $	40.77%

Finally, in the menu bar in the *File* area, choose Save  icon again and then **Save** to save your changed story.

Save

Step 14: Ranking Lists

Task Create two ranked lists. **Time** 10 min.

Description Create a Top 5 and a Bottom 5 ranking list based on the sold quantities on the *Product View* tab page.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Ranked Lists

Ranked lists, also known as conditions, are used to display top and top lists. Bottom values. They consist of a condition and a corresponding threshold value.

Ranking lists are sorted according to the condition so that the statements can also be captured visually quickly.

For decision support, the Top 5 and Bottom 5 bicycles, measured by their quantity sold, are to be displayed as a bar chart on the *Product View* tab.

Switch to *"Product View"* tab page. The charts are to be inserted to the right of the *heat map*.

More Actions → Add Lane → Add Lane to Right

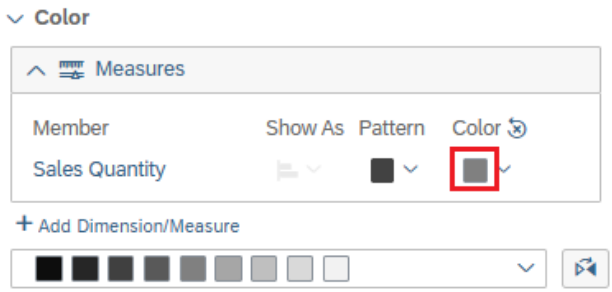
On the *Product View* page, first insert a new area next to the *heat map* area. To do this, click the heat *map* area and choose **More Actions** ( ) ▶ **Add Lane** ▶ **Add Lane to Right.**

Chart

In the new area, use the ribbon to insert a graphic in the *Insert* area by clicking the **Chart** icon .

Sales Quantity
Product
Grayscale
medium grey tone

The right side panel opens. In the Builder, define ***Sales Quantity*** as the *measure*, **Product** as the *dimension*, **grayscale** as the color scheme, and select a **medium gray tone** as the *color.*

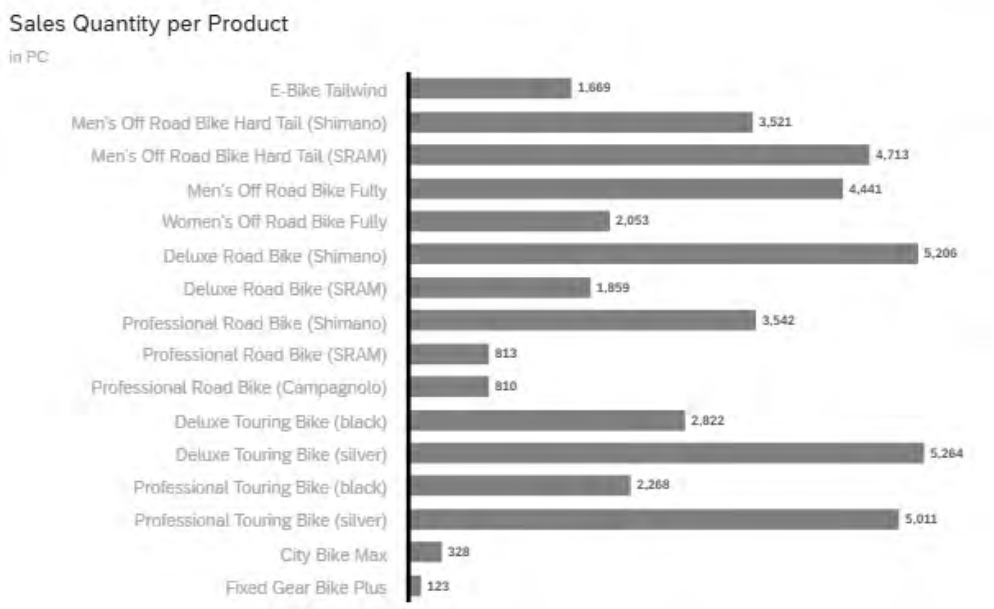


Bicycles

To only evaluate bicycles in the new chart, you then define a corresponding filter on Bicycles on *Division* (under *Product*), which cannot be changed by the end user.

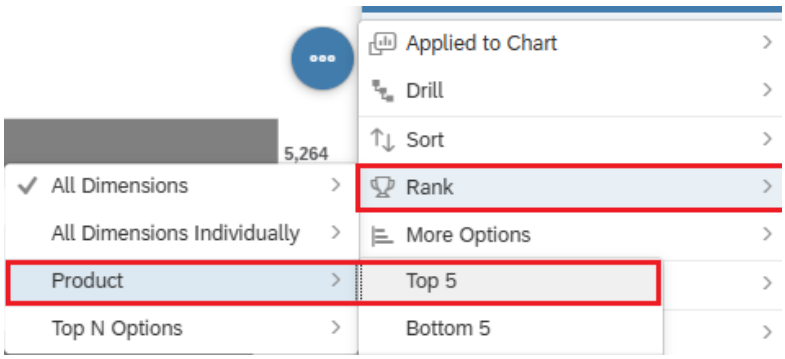
Level 3

Since *Product* is not to be displayed as a hierarchy, but as a flat list, now use the mouse to move to the area of the *Dimension Product* and select **level 3** after clicking on the symbol (*Hierarchy*).



More Actions ->Rank -> Product -> Top 5

Click on the area of the graphic and choose under **More Actions** **Rank ► Product ► Top 5**.



Top 5 Bicycles by Quantity

Finally, change the title of the graphic to **Top 5 Bicycles by Quantity**. To do this, double-click the title text and adjust it accordingly.

Styling

In the *right-hand side panel*, switch from the *Builder* panel to the **Styling** panel.

Fullscreen

In the *Quick Menus* area, deselect all options except **fullscreen**.

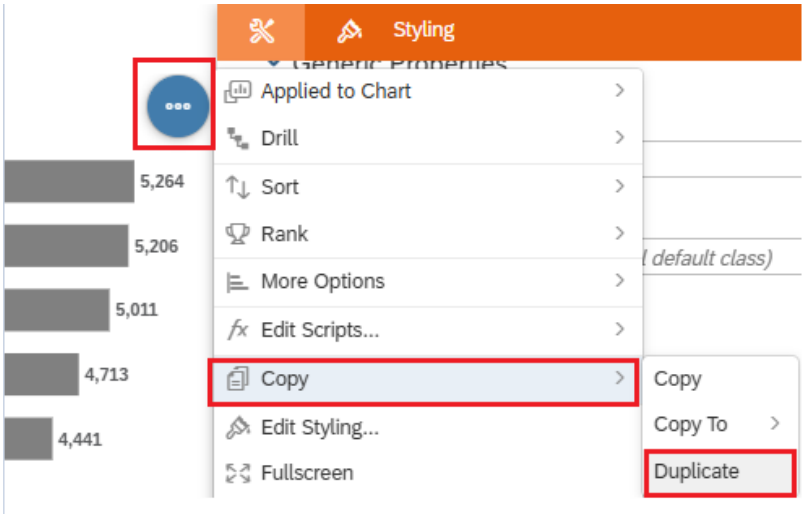
Height 13 Grid
Width 12 Grid

Change the size of the graphic by changing the *Height* to **13 Grid** and *width* to **12 Grid** under *Size and Position*.

Since the second chart for the Bottom 5 bikes is to be displayed in the same style, copy the Top 5 chart and change it to a Bottom 5 chart.

When you click on your diagram, the option More Actions appears in the upper right corner and follows the path: **Copy ► Duplicate**.

Copy -> Duplicate

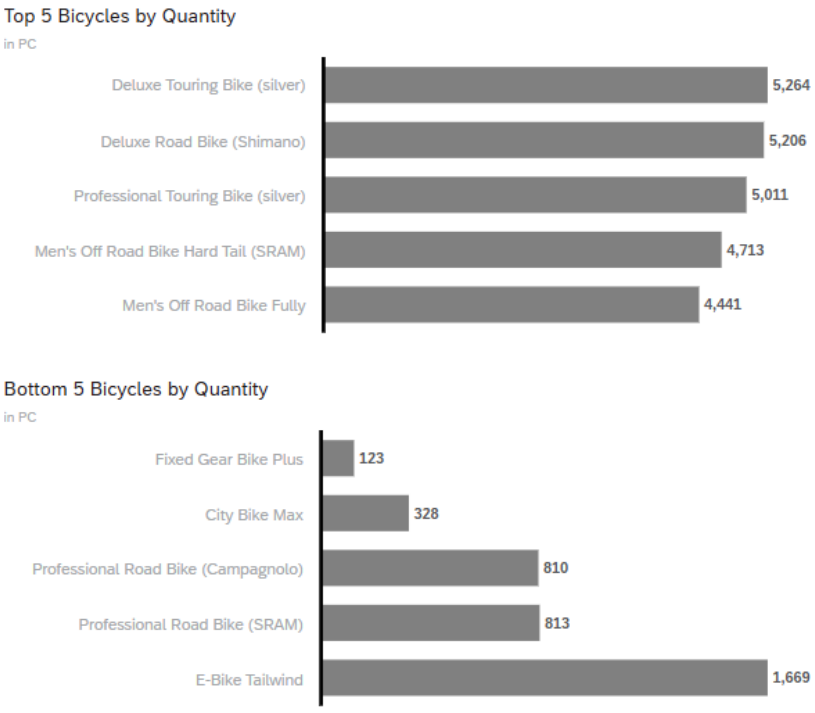


The second chart is now inserted below your Top 5 chart. Change the ranking of the second chart to *Bottom 5*. When you click on your graphic, **More Actions ► Rank ► Product ► Bottom 5** appears at the top right.

More Actions → Rank → Product → Bottom 55

Finally, change the title of the chart to **Bottom 5 Bicycles by Quantity**.

Bottom 5 Bicycles by Quantity



Finally, in the menu bar in the *File* area, choose Save icon again and then **Save** to save your changed story.

Save

Step 15: Variances

Task Create time variances. **Time** 10 min.

Description Create various time differences in graphics on the *Product View* and *Dynamic Analysis* tabs.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Variances

Differences between time dimensions provide the simple option of differences with red (negative difference) or Display green (positive difference) without programming knowledge or similar.

Variances are available if a time dimension is available in your model.

You want to set two time variances in your story. On the one hand, on the *Product View* tab in the *Top* or *Bottom 5 Bicycles by Quantity* and in the chart on the *Dynamic Analysis* tab.

Start with the time variance on the *Product View* tab and in the *Top 5 Bicycles by Quantity* chart.

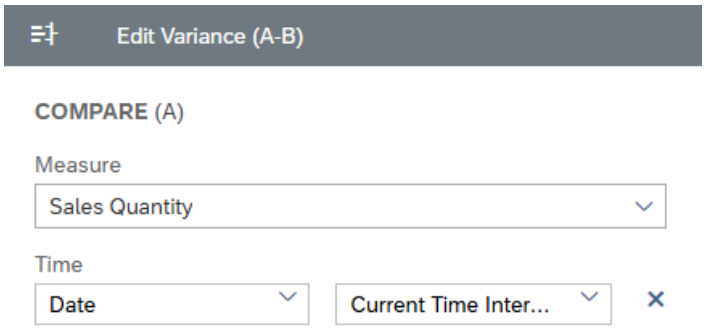
Open the *right side panel* in the *Builder* panel and click in the *Top 5 Bicycles by Quantity* chart.

Chart Add-Ons
Variance…

Navigate to the **Chart Add-Ons** area and then choose **Variance…**.

Sales Quantity
Add Version/Time
Date
Current Time Interval

The definition view opens in the right-hand pane. First, in the *Compare (A)* area, under *Measure*, select the Measure **Sales Quantity**. Then choose **Add Version/ Time**. After you have selected **Date**, select **Current Time Interval**.



Sales Quantity
Date Previous Year

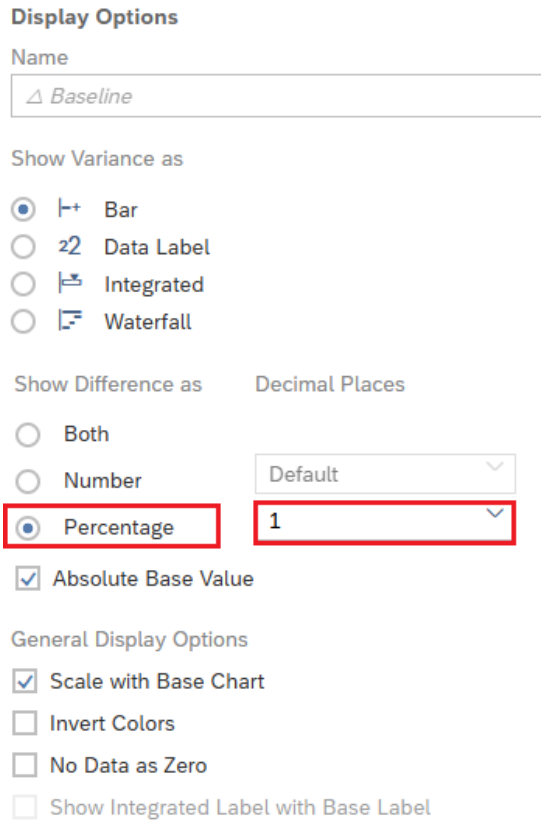
In the *Compare (B)* area, define the measure **Sales Quantity** under *Measure*. Now select **Previous Year** as *the Date*.

Percentage

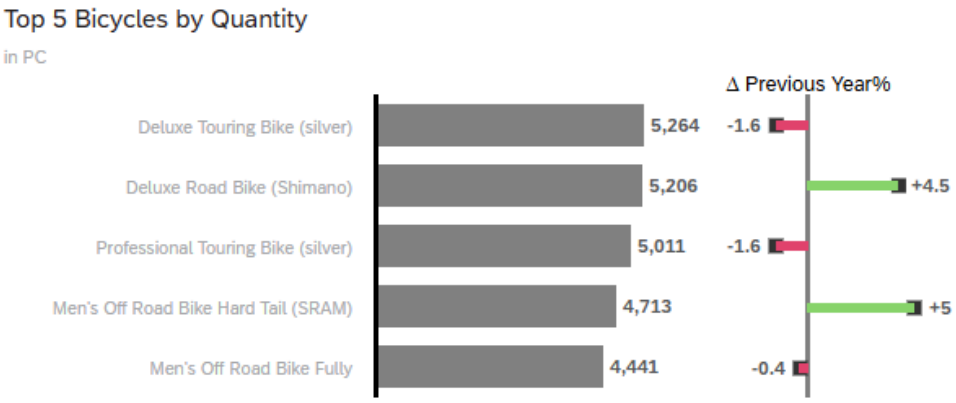
In the *Display Options* area, under *Show Difference as*, select the value **Percentage** to display the variance as a percentage value rather than an absolute value.

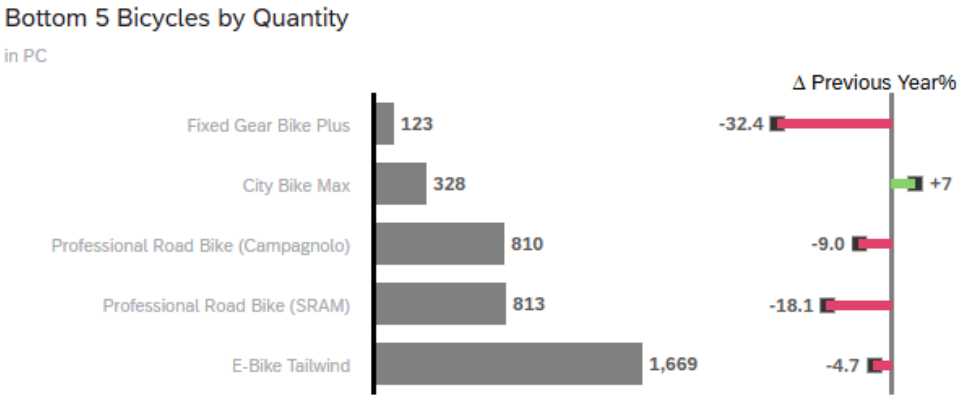
Set the *Decimal Places* to **1** and finish the definition with **Done**.

1
Done



Do the same for the *Bottom 5 Bicycles by Quantity* graphic.





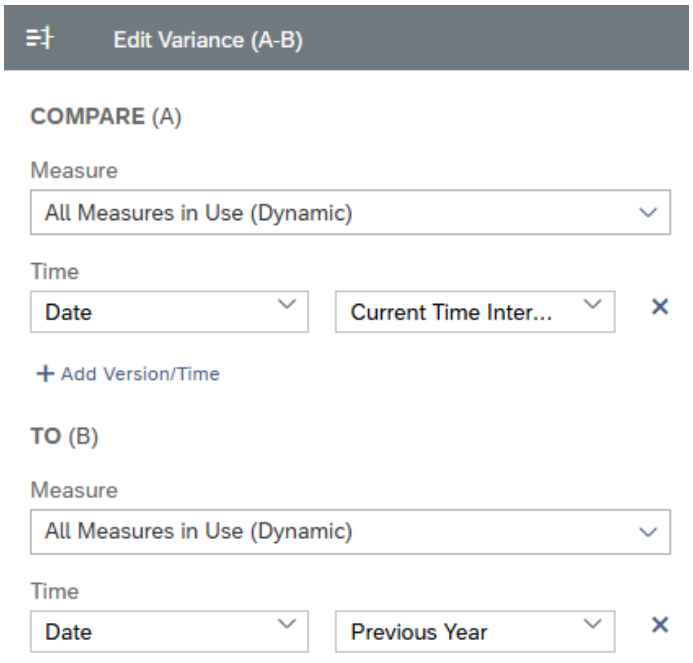
Before you set the time difference on the *Dynamic Analysis* tab, choose the *Save* icon in the *File* area of the ribbon and choose **Save** to save your changed story.

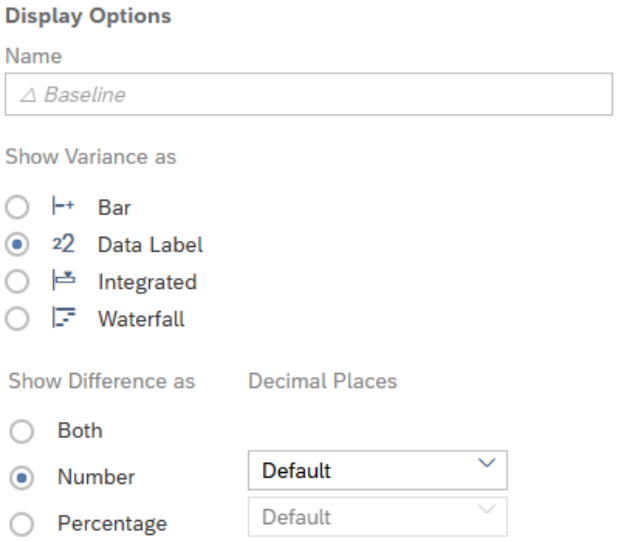
Save

Switch to the *Dynamic Analysis* tab and also set a time variance *Previous Year*. To do this, proceed as for the Chart *Top 5 Bicycles by Quantity.*

Leave the *Measure* set to **All Measures in Use (Dynamic)** and the Display to **Number** without *Decimal Places*. In the *Display Options* area, under *Show Variances as*, select **Data Label** and complete the definition with **Done**.

Display Options ->
Show Difference as
Percentage
~~Numeric~~
Ok then

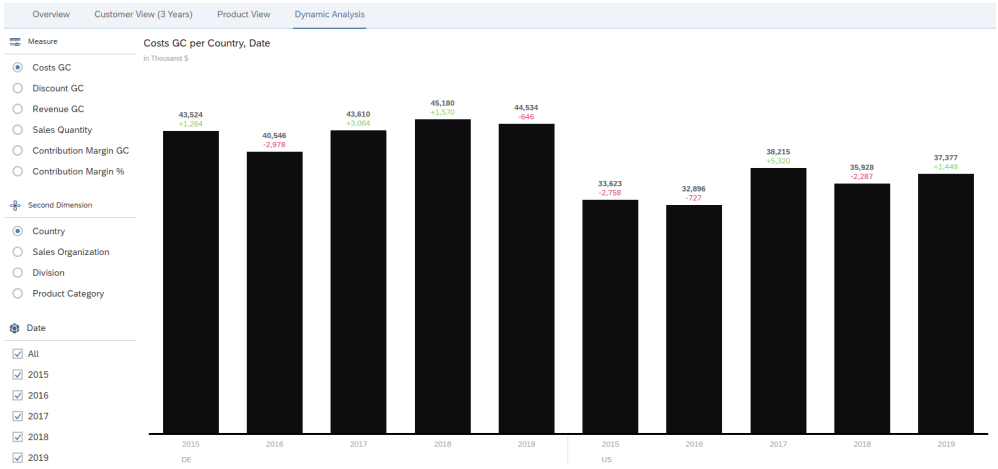




In the *Measure* selection filter, set the selection option for the contribution *margin GC* and *contribution margin %* measures. To do this, right-click in the *Measure* selection filter, click **Edit Input Control**, and select the key figures **Contribution Margin GC** and **Contribution Margin %**.

Edit Input Control
Contribution Margin GC
Contribution Margin %

Now adjust the size and position of the elements so that all key figures can be selected directly using the radio button.



Finally, in the menu bar in the *File* area, choose Save  icon again and then **Save** to save your changed story.

Save

Step 16: Graphic Numerical Point

Task Create multiple numeric point graphics. **Time** 10 min.

Description Create four graphics of type *Numeric Point* incl. Previous year's variances on the "Overview" tab page.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Numeric Point

Numeric point graphics are key figure graphics that enable you to highlight the most important key figures and provide them with different comparisons (for example, for the plan, for the previous month, for the previous year).

For the year 2019, the key figures *Revenue GC, Sales Quantity, Costs GC*, and *Contribution Margin %* are to be mapped as *numerical point* graphics including a comparison of the previous year.

More Actions → Add Lane → Add Lane Above

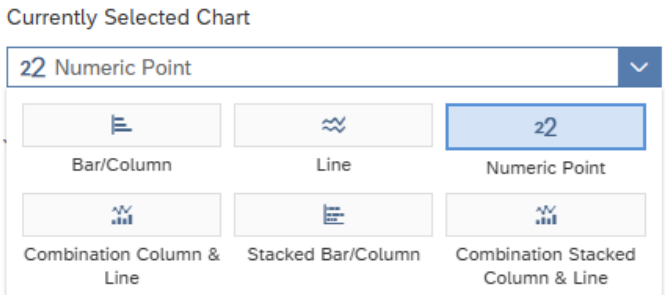
Switch to the *Overview* tab page. For the graphics, insert a new area above the time series area. To do this, click the time series area and choose **More Actions (…) ► Add Lane ► Add Lane Above.**

Chart

Use the ribbon to insert a graphic in the *Insert* area by clicking the **Chart** icon.

Current Selected Chart Numeric Point

The right side panel has opened. In the business area, change the graphic type to *Numeric Point*. To do so, open the dropdown under **Current Selected Chart** and select the **Numeric Point entry.**



Make the following settings for your graphic.

Revenue GC

Define **Revenue GC** as primary *value*.

2019

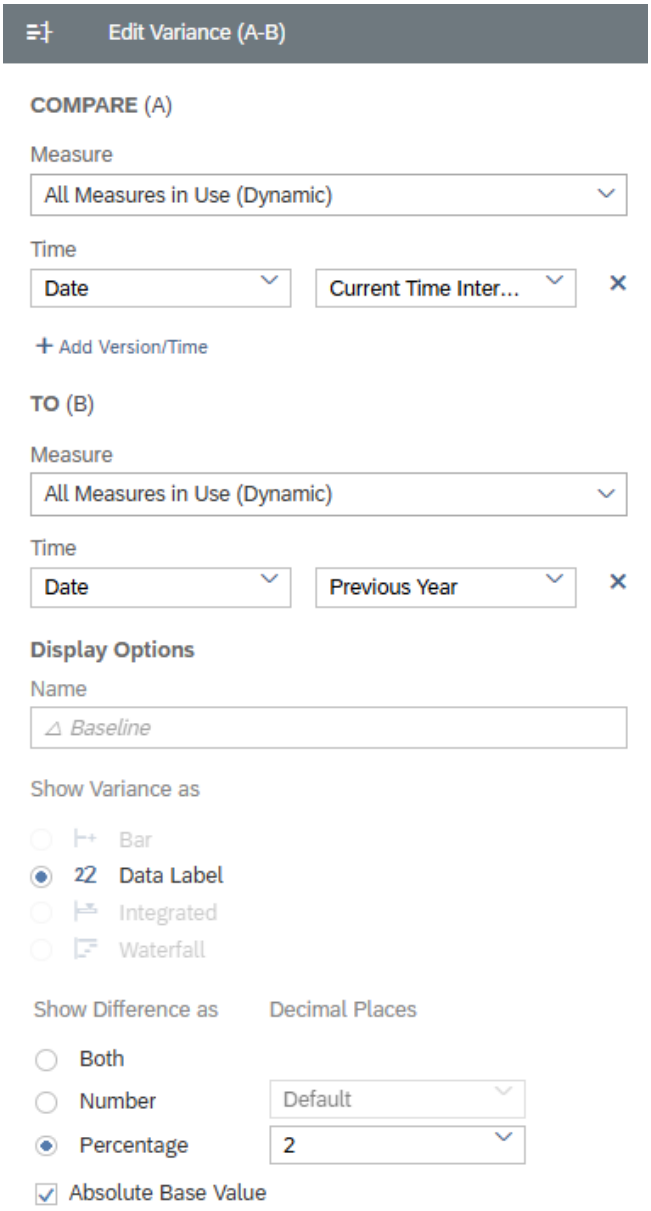
For the Date dimension, you define a filter for **2019**, which cannot be changed by the end user.

Chart Add-Ons Variance…

Open the **Chart Add-Ons** area and then choose **Variance…**.

Leave the *measure* as **All Measures in Use (Dynamic)** and choose **Add Version/ Time**. After you have selected **Date**, select **Current Time Interval**. For the comparison, you define the *Time* as **Previous Year**. Finally, under *Display Options*, you define the *difference* as **a percentage** with **2** decimal places. Choose **Done** to complete the definition.

All Measures in Use (Dynamic)
Add Version/Time
Date
Current Time Interval
Previous Year
Percentage
2
Done



Disable the representation of all *subtitles* and the primary value label. When you click on your graphic, the **option More Actions** ( ) appears in the upper right corner. Click the button. In the context menu, choose **More Options ► Show/Hide ►Subtitle ► All** and **Primary Value Labels.**

More Actions
More Options →
Show/Hide →
Subtitle →
~~All~~
~~Primary Value Labels~~

In the *right-hand side panel*, switch from the *Builder* panel to the **Styling** panel.

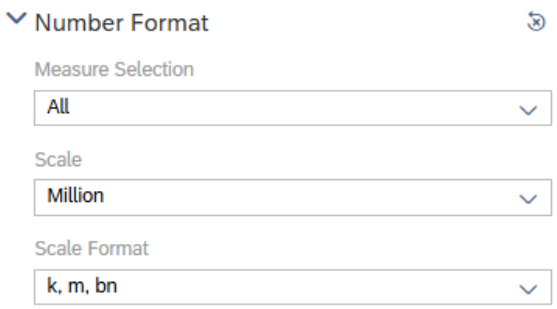
Styling

In view mode, the users should not have a context menu available. In the *Quick Menus* area on the *right-hand side* of the screen, deselect it.

~~Quick Menus~~

Now scroll down to the *Number Format* area. Select *Scale* as **Million** and *Scale Format* as **k, m, bn**.

Million
k, m, bn



In the *Font* area under *Text Selection*, select the listed elements one after the other and change the font settings as specified:

- **Primary** → *Alignment* **Center Text**
- **Primary Value** (155,92) → *Size* **60**; *Font Style* **Bold**
- **Primary Variance Value** (+7.33%) → *Size* **28**
- **Primary Variance Title** (2018) → *Size* **18**

Primary → Center Text
Primary Value → 60, Bold
Primary Variance Value → 28
Primary Variance Title → 18

Change the size of the graphic under *Size and Position* at *Height* to **8 Grid**.

8 Grid

Finally, change the title to **Revenue GC 2019**.

Revenue USD



To create the three other graphics, you want to copy the graphics you just created and change the key figures accordingly.

First, you insert 3 more lanes. To do this, click on the area of the graphic and choose **More Actions** ► **Add Lane** ► **Add Lane to Right.**

More Actions →
Add Lane →
Add Lane to Right

Click the area of the graphic and choose **More Actions** ► **Copy** ► **Copy**.

More Actions →
Copy →
Copy

To insert the copied graphic, click in the empty area and press **CTRL + V** on your keyboard.

CTRL + V

Perform these steps 2 more times and then change the settings and titles of the graphics as follows:

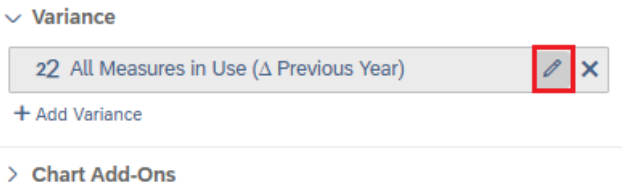
- Figure 2: **Sales Quantity,** *Scale->* **Thousand**
- Figure 3: **Costs GC**
- Figure 4: **Contribution Margin %**

Sales Quantity
Thousand
Costs GC
Contribution Margin %

Since, in contrast to revenue (*revenue GC*), an increase in the cost of goods manufactured (*costs GC*) is probably negative, invert the color display of the cost of goods manufactured.

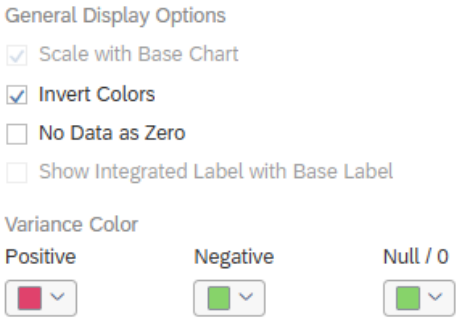
To do this, select the 3 graphic. On the *right-hand side panel* in the *Builder* panel, scroll to the *Variance* area and choose **Edit** .

Edit



In the *Edit Variances (A-B)* menu that opens, in the *Display Options* area, under *General Display Options*, select the **Invert Colors** setting and choose **Done**.

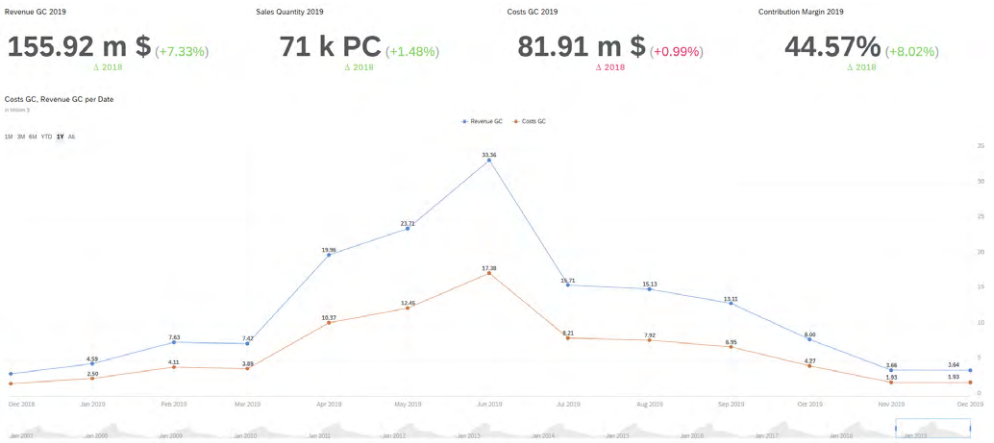
Invert Colors
Done



Rename your new graphics to a meaningful name and adjust the height of the time series at *Height* to **33 Grid**.

33 Grid

As a result, your page should look as follows.



Finally, in the menu bar in the *File* area, choose Save icon again and then **Save** to save your changed story.

Save

Step 17: Calculated Characteristics

Task Create a calculated characteristic. **Time** 10 min.

Description Create a calculated characteristic based on the product category to create a clear heat map.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Calculated Characteristics

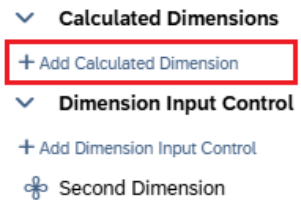
Calculated *characteristics* can be used to format text content, among other things. It is possible to remove fixed text parts or link several texts to a new element.

On the *Product View* page, the Product *Category* is currently displayed on the x-axis in the *Heat Map*. Since the values for the *Division Bicycles* always contain the addition *Bikes*, this should be removed so that the label is not unnecessarily wider.

Switch to the *"Product View"* tab page. Open the *right side panel* in the *Builder* panel. Click into the *Heat Map Quantity for Bicycles*.

~~Product Category~~
At least 1 dimension required
Add Calculated Dimension

Remove the **Product Category** in the *X-Axis* area and then choose **At least 1 Dimension required** and choose **Add Calculated Dimension**.

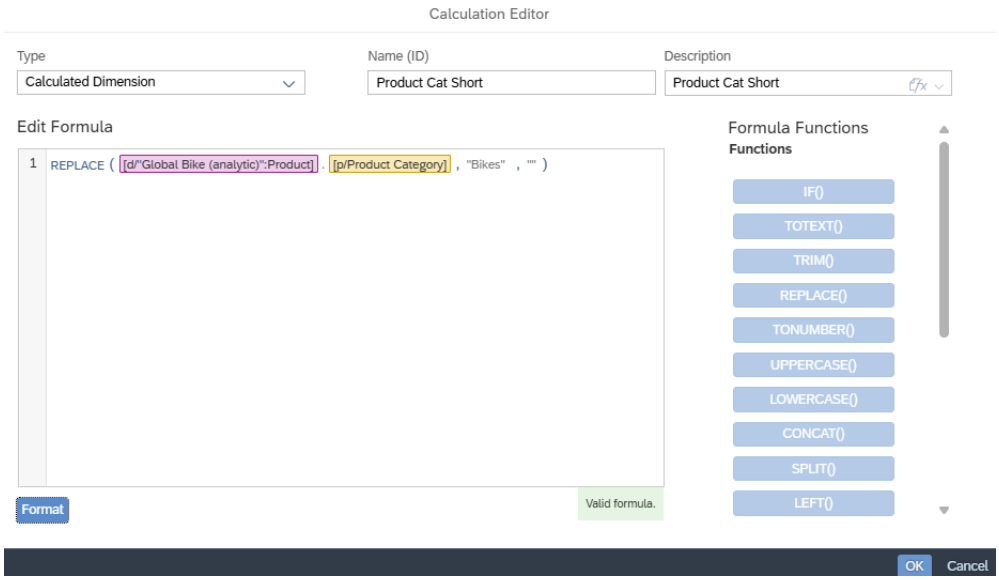


Calculated Dimension
Product Cat Short

In the *calculation editor* that opens, first select **Calculated Dimension** as the type. Enter Product **Cat Short** as the *Name* and *Description*.

Replace
CTRL + Space
Ok then

Use the formula `REPLACE ( [d/"Global Bike (analytic, measure)":Product] . [p/Product Category] , "Bikes" , "" )` to replace *"Bikes"* with*""*. [p/…] can only be selected with *CTRL + Space* or *Alt Gr + square bracket*. When you have completely defined your dimension, confirm the window with **OK**.



Now adjust the sizes of the *heat map* and *table*.

To do so, go to the **Styling** panel on the *right-hand side* panel from the *Builder* panel.

Styling

Click in the *Heat Map* and under *Size and Position*, change the *widget* to **7 Grid** and *Height* to **26 Grid**. Change the value under *Left* to **0 Grid** so that the graphic on the left starts with the area.

To give the end user of your story a quick overview of the totals, you now set a totals row for *Product Cat Short*.

Now reduce the heat *map* area to the size of the graphic on the left-hand side. This increases the space for the table.

Click in the *table* and under *Size and Position*, change the *widget* to **17 Grid** and *Height* to **26 Grid**.

17 Grid
26 Grid

Now click in the *table* in the header *Contribution Margin GC* and now scroll in the *right-hand side area* to the *area Number Format*. Under *Decimal Places*, select **0**.

0



Bicycles

Date	Product	Measures	Sales Quantity	Contribution Margin GC	Contribution Margin %
(all)	(all)		44,443 PC	68,826,302 $	44.48%
2019	(all)		44,443 PC	68,826,302 $	44.48%
	Bicycles		44,443 PC	68,826,302 $	44.48%
	E Bikes		1,669 PC	3,475,673 $	39.27%
	Offroad Bikes		14,728 PC	18,276,925 $	47.09%
	Road Bikes		12,230 PC	19,885,494 $	45.53%
	Touring Bikes		15,365 PC	26,843,992 $	42.92%
	Trend Bikes		451 PC	344,217 $	40.77%

Quantity for Bicycles

in PC

	E	Offroad	Road	Touring	Trend
Jan	39	359	340	584	5
Feb	67	837	603	761	20
Mar	69	696	581	750	10
Apr	178	1,874	1,715	1,831	67
May	330	2,213	1,874	2,183	73
Jun	368	3,170	2,485	3,364	85
Jul	164	1,457	1,193	1,656	5
Aug	127	1,469	1,212	1,401	61
Sep	149	1,097	937	1,467	44
Oct	83	737	685	785	20
Nov	30	409	310	310	17
Dec	65	410	295	273	44

Finally, in the menu bar in the *File* area, choose Save icon again and then **Save** to save your changed story.

Save

Step 18: Measure Filtering

Task Create a measure filtering. **Time** 10 min.

Description Create a measure filter to represent those bikes that have a sales quantity of at least 2000 pieces.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Measure Filtering

You can use *key figure filtering* to evaluate data that exceeds or falls short of a defined value of a key figure. In this form of filtering, note that the characteristics for which the key figure is to exceed or fall below a defined value must be defined.

A new tab will list the bikes for which a minimum sales volume of 2000 pieces was achieved in 2019. The display should break down the results by sales organizations.

Most Popular Bikes

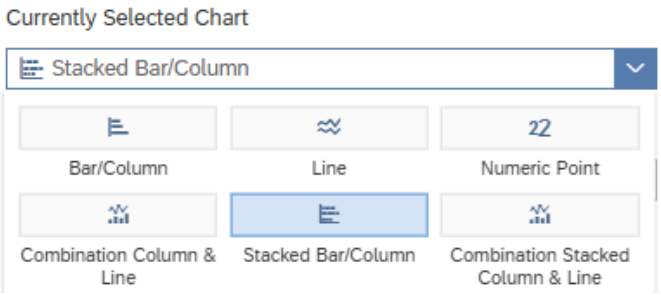
To do this, you define the *Most Popular Bikes* page by creating a new *Responsive* page and renaming it **to Most Popular Bikes**.

Insert -> Chart

Use the ribbon to insert a graphic in the *Insert* area by clicking the **Chart** icon.

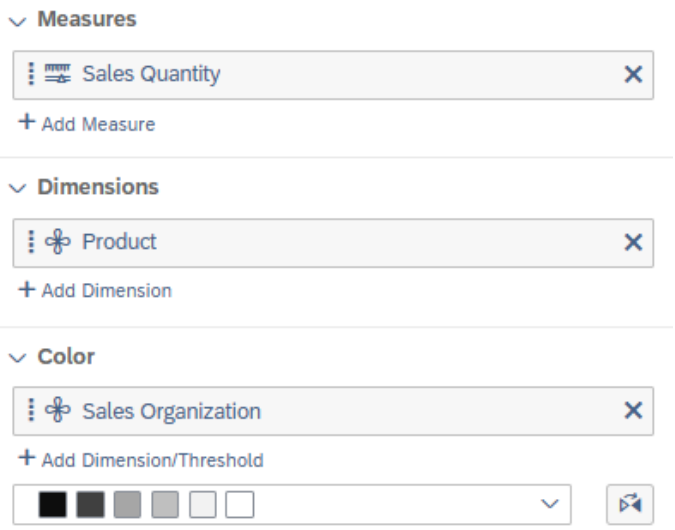
Current Selected Chart
Stacked Bar/Column

The right side panel is open. In the *Builder* panel, change the chart type to *Stacked Bar/Column.* To do this, open the dropdown under **Current Selected Chart** and choose the entry **Stacked Bar/Column.**



Sales Quantity
Product (Level 4)
Sales Organization
Grayscale

Define **Sales Quantity** as *Measure*, **Product (Level 4)** as *Dimension*, and **Sales Organization** as *Color*, and **Grayscale** (*4 Colors*) as *Color Scheme.*



Sales Organization
Grayscale

For the division dimension, you define a filter on **bicycles** that cannot be changed by the end user.

Bicycles

You also insert a new area above the area of the graphic. To do this, click on the area of the graphic and choose **More Actions** ► **Add Lane** ► **Add Lane Above.**

More Actions →
Add Lane →
Add Lane Above

Now define a *Page Filter* on *Date*, with which the end users can select a maximum of the last 3 years.

To do this, click in the new area and insert a filter using the *Insert* ribbon and click on the icon **input control** .

Input Control

The system then inserts a *Page Filter* element. Choose the following path: **Dimensions** ► **Date** ► **Filter by Member…**.

Dimensions →
Date →
Filter by Member…

In the following dialog, select the values for **2017 - 2019** individually. Since only a single selection of the year is to be possible in this filter, choose the **Settings** button and define the *selection type* as **Single Selection**. Confirm the window with **OK.**

2017 – 2019
Settings
Single Selection
Ok then

You also want to insert a key figure filter as a *page filter* so that the end users can lower limits.

To do so, add another filter. In the menu that opens, select **Measures** ► **Sales Quantity**.

Measures →
Sales Quantity

In the following dialog, you first define the dimensions for determining the key figures. Select Below *Dimension Context* **Year** (below *Date*) and **Product**.

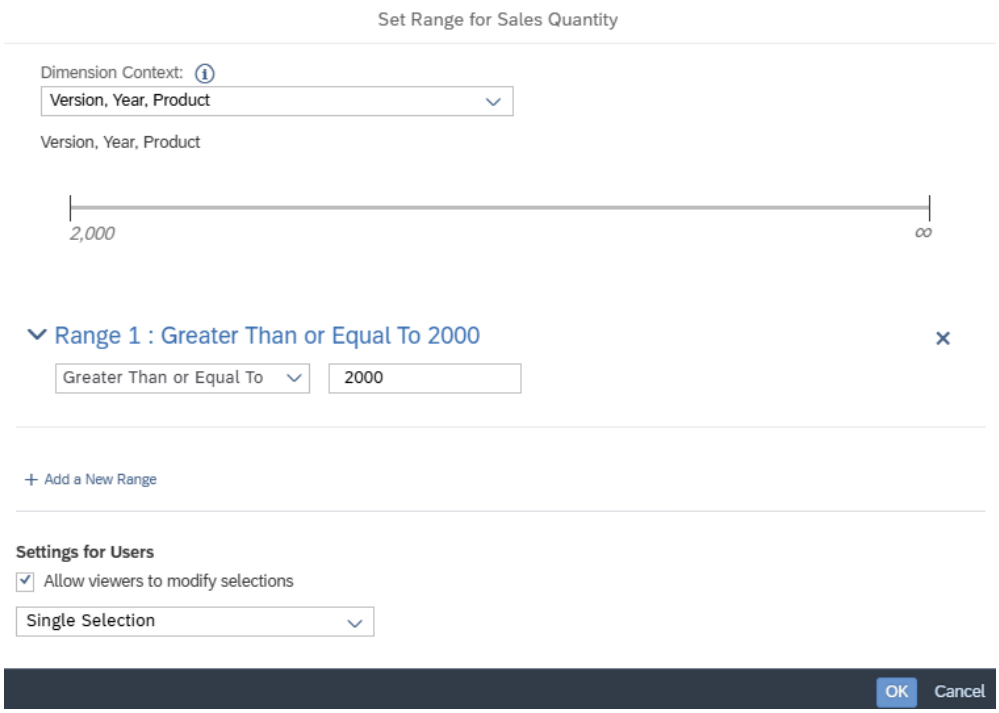
Year
Product

Under *Range*, select **Greater Than or Equal To** and enter **2000** as the value.

Greater than or equal to
2000

Allow end users to change the value as **Single Selection**. Then confirm the window with **OK**.

Single Selection
Ok then



In the *right-hand side panel*, switch from the *Builder* panel to the **Styling** panel.

Styling

Change the size of the *Page Filter* for *Date* by clicking it and changing the *Height* to **7 Grid** under *Size and Position*.

Height 7 Grid

You change the size of the *key figure filter* to **5 Grid** for *Width* and **7 Grid** for *Height*.

Width 5 Grid
Height 7 Grid

In the *Quick Menus* area, deselect it for both filter elements.

~~Quick Menus~~

For the filters on *Date*, deactivate the *Cascading Effect* button by right-clicking in the filter and selecting **Cascading Effect**.

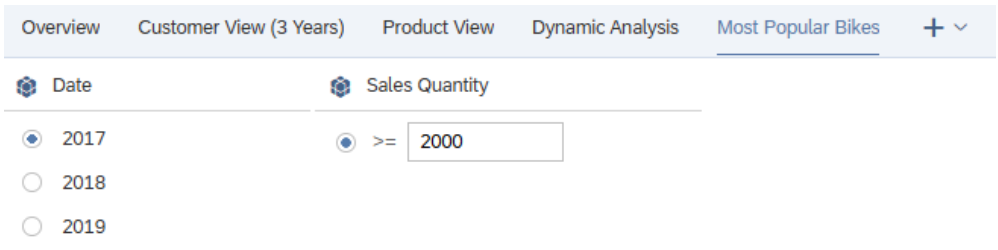
~~Cascading Effect~~

For the key figure filter, deactivate the *range description* by choosing **Show/Hide ► Range Description**.

Show/Hide →
~~Range Description~~

Now change the name of the measure filter to **Sales Quantity**.

Sales Quantity



To display the highest-selling bike in the first line, click on the graphic and choose **More Actions** ▶ **Sort** ▶ **Sales Quantity** ▶ **Highest to Lowest**.

More Actions →
Sort →
Sales Quantity →
Highest to Lowest

In the *Quick Menus* area on the *right-hand side* of the screen, deselect all options except **fullscreen**.

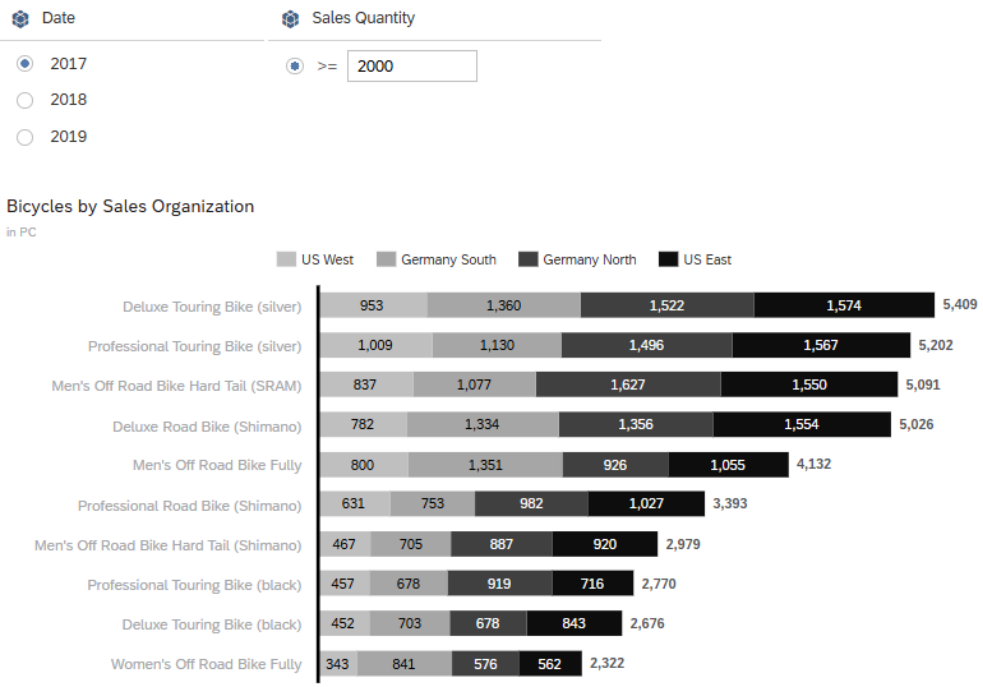
Fullscreen

Rename the graphic to **Bicycles by Sales Organization**.

Bicycles by Sales Organization

Under *Styling, Size and Position,* adjust the size of the graphic to **Width 15** and **Hight 20**.

Width 15 Grid
Height 20 Grid



Finally, in the menu bar in the *File* area, choose Save icon again and then **Save** to save your changed story.

Save

Step 19: Linked Analysis

Task Create linked analysis **Time** 10 min.

Description On the *Most Popular Bikes* tab, create a *linked analysis* with a 2. Graphic of the type *combined graphic*.

Initial Screen

You are still in the *story Story_Case_Study* from the last step. If you have closed them, search for them and find them in your personal folder. Make sure you are in *Edit Mode*.

Measure Filtering

Linked analysis can be used to filter further elements using the selection in a graphic. It is possible to select different elements in a graphic to use these values as filtering.

Note that all linked *analysis* elements must be based on the same model.

On the *Most Popular Bikes* tab, you want to show the relationship between the key figures *Sales Quantity* and *Contribution Margin %* over the course of the months.

More Actions → Add Lane → Add Lane to Right

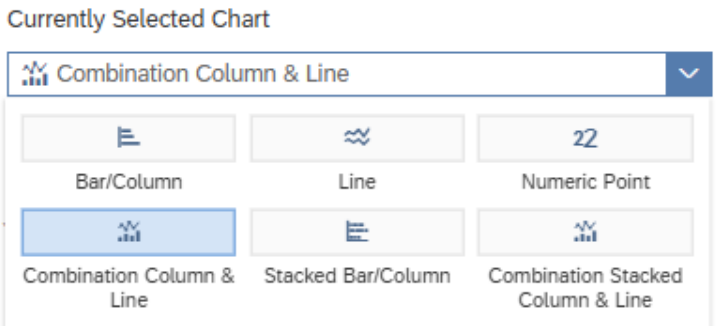
Switch to the *Most Popular Bikes* tab and insert a new section to the right of the existing graphic. To do this, click the area of the graphic and choose **More Actions** ( ) ► **Add Lane** ► **Add Lane to Right.**

Insert -> Chart

Use the ribbon to insert a graphic in the *Insert* area by clicking the **Chart** icon.

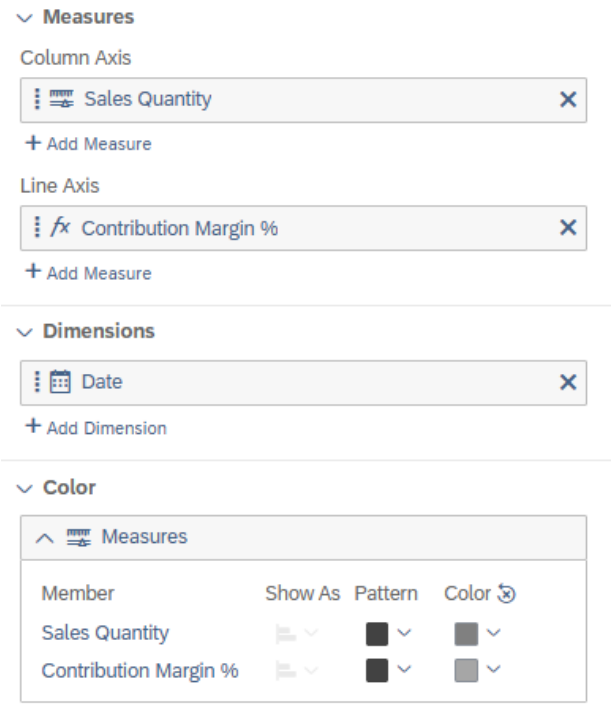
Current Selected Chart Combined Column & Line

The right side panel is open. In the *Builder* panel, change the chart type to *Combined Column & Line*. To do this, open the dropdown under **Current Selected Chart** and choose the combined **column & line entry.**



Sales Quantity
Contribution Margin %
Date (Level 3)
Grayscale

Define the key figure **Sales Quantity** as *column axis*, the key figure **Contribution Margin %** as *line axis*, **Date (Level 3)** in the dimensions area and **grayscale** (*4 colors*) as *color scheme.*

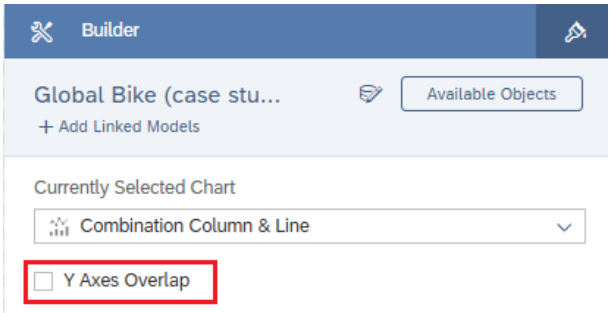


Now define *a filter for the division dimension* on **bicycles**, which cannot be changed by the end users.

Bicycles

Deactivate the link between the two key figures in order to be able to analyze the histories individually, since there is no connection between them. To do this, disable **Y Axes Overlap**.

Y Axes Overlap



In the *right-hand side panel*, switch from the *Builder* panel to the **Styling** panel.

Styling

In the *Quick Menus* area on the *right-hand side* of the screen, deselect all options except **fullscreen**.

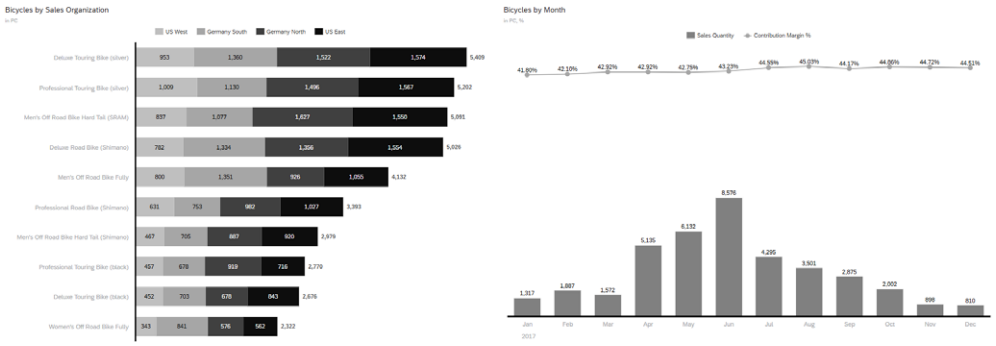
Fullscreen

You now adjust the heading of the graphic. To do this, double-click in the text and enter **Bicycles by Month**.

Bicycles by Month

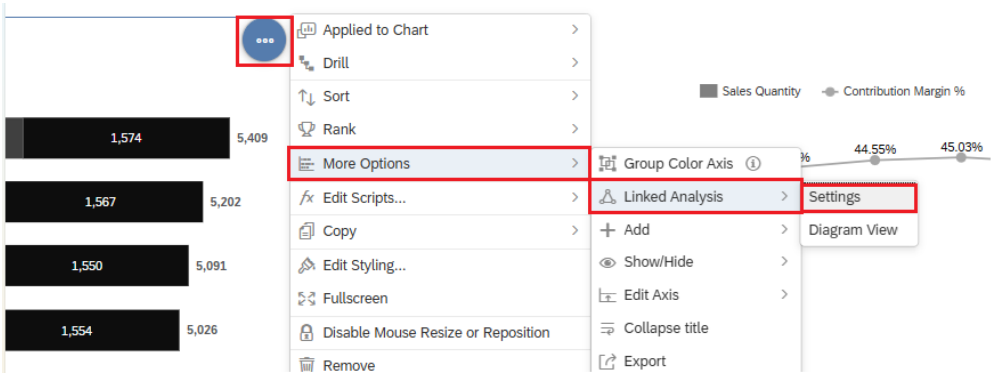
Adjust the size of both graphics under *Size and Position* to **18 Grid** for *Widht* and **32 Grid** for *Height*.

Widht 18 Grid
Height 32 Grid



Next, you define the *linked analysis* for the *left graphic*. If you click on the left graphic, the option **More Actions** appears at the top right. Click the button. In the context menu, choose **More Options ▶ Linked Analysis ▶ Settings**.

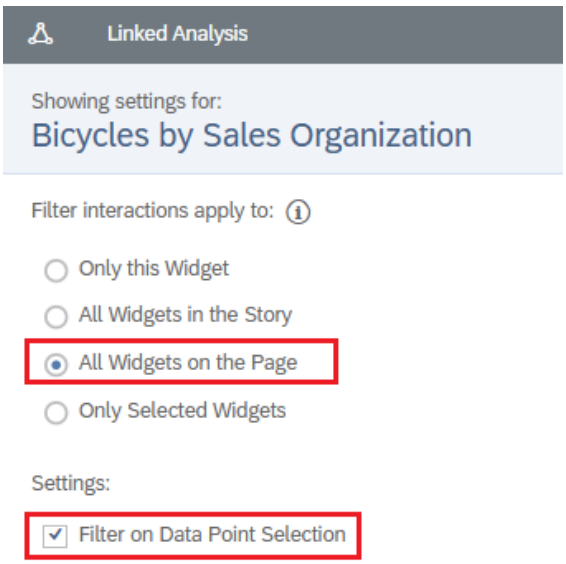
More Actions
More Options → Linked Analysis → Settings



The definition view now opens in the right-hand pane. First, in the Filter *Interactions apply to:* area, select **All Widgets on the Page**. In the *Settings* area that then appears, choose the **Filter on Data Point Selection** option and close the window by **choosing Apply**.

All Widgets on the Page

Filter on Data Point Selection
Apply

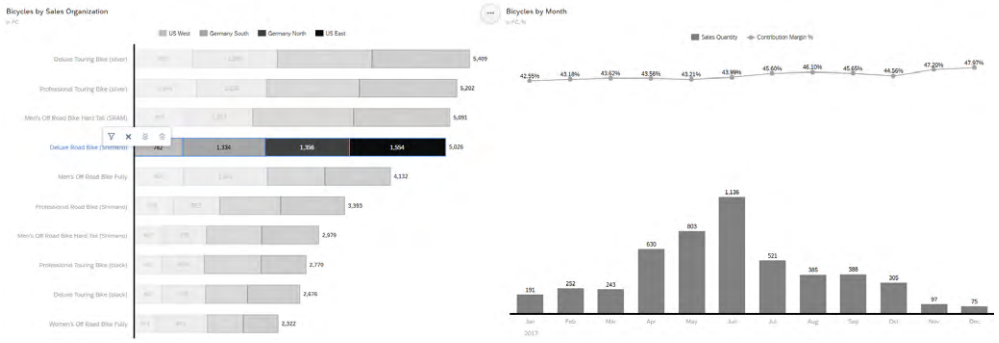


Now proceed in the same way as in the *graphic on the right*.

Save

Finally, in the menu bar in the *File* area, choose Save  icon again and then **Save** to save your changed story.

Test your link, for example, by selecting a row or a subbar in the stacked graphic.



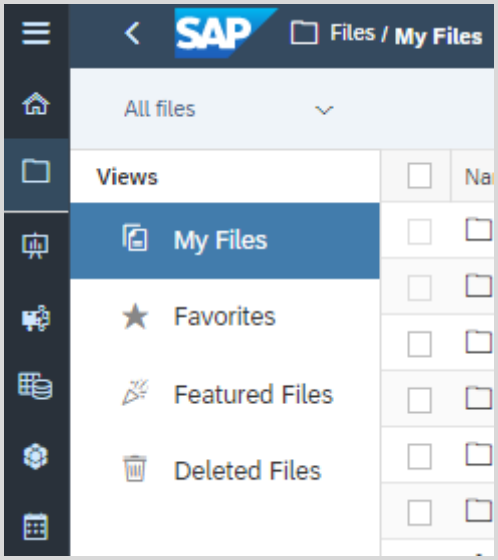
Step 20: [OPTIONAL] Share the story with instructor

Task Share your story with your instructor. **Time** 5 min.

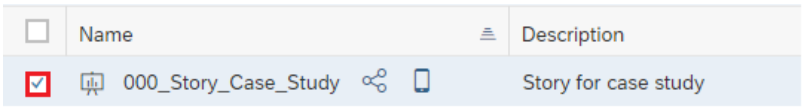
Create a copy of your story and place it in the folder named by your instructor.

Initial Screen

On the home screen, click (*Files*) in the upper left corner.



Navigate to your personal folder where you saved the story from this case study, probably *My Files* ► *My SAC course* ► *Stories*. Select your story by selecting the box at the beginning of the row.

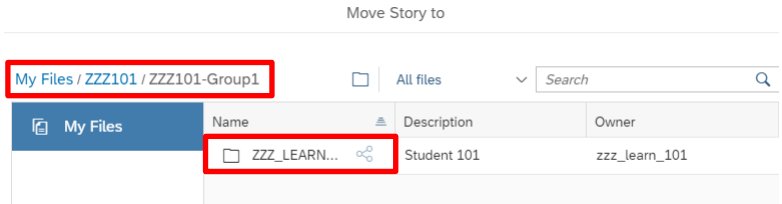


Copy To

Now click the **Copy To** icon in the menu bar.

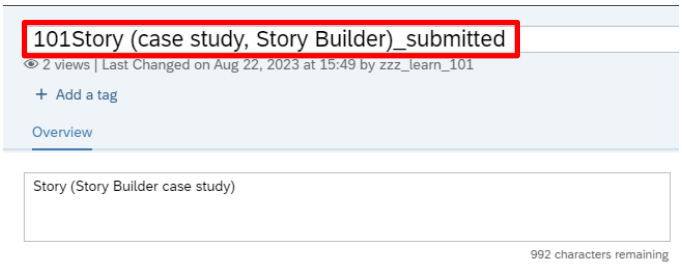


In the window that opens, select a folder to which only your instructor and you have access. If necessary, ask your instructor.



To be able to later distinguish the copied story from the original, add the suffix "**_submitted**" to the *Name* field.

_submitted



Confirm your entry with **OK** and inform your instructor.

Ok then